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Debiasing Session-Based Recommendation for the Digital Economy: Propensity-Aware Training and Temporal Contrast on Graph Transformers

Yongjian Wang [1], Junru Si [2], Xuhua Qiu [2,*] and Kunjie Zhu [2,3,*]

[1] School of Foreign Languages, Guangdong Polytechnic Normal University, Guangzhou 510640, China; wangyongjian@gpnu.edu.cn
[2] Institutes of Science and Development, Chinese Academy of Sciences, Beijing 100190, China; sijunru25@mails.ucas.ac.cn
[3] Department of Mathematics, City University of Hong Kong, Hong Kong, China
* Correspondence: qiuxuhua23@mails.ucas.ac.cn (X.Q.); kunjie.zhu@my.cityu.edu.hk (K.Z.)

Abstract

Session-based recommender systems (SBRs) are critically impaired by exposure bias in observational training logs, causing models to overfit to logging policies rather than true user preferences. This bias distorts offline evaluation and harms generalization, particularly for long-tail items. To address this, we propose the Propensity- and Temporal-consistency Enhanced Graph Transformer (PTE-GT), a principled framework that enhances a recent interval-aware graph transformer backbone with two synergistic training-time modules. This Graph Neural Network -based architecture is adept at modeling the complex, graph-structured nature of session data, capturing intricate item transitions that sequential models might miss. First, we introduce a propensity-aware (PA) optimization objective based on the self-normalized inverse propensity scoring (SNIPS) estimator. This module leverages logs containing randomized exposure or logged behavior-policy propensities to learn an unbiased risk estimate, correcting for the biased data distribution. Second, we design a lightweight, view-free temporal consistency (TC) contrastive regularizer that enforces alignment between session prefixes and suffixes, improving representation robustness without computationally expensive graph augmentations, which are often a bottleneck for graph-based contrastive methods. We conduct comprehensive evaluations on three public session-based benchmarks—KuaiRand, the OTTO e-commerce challenge dataset (OTTO), and the YOOCHOOSE-1/64 split (YOOCHOOSE)—and additionally on the publicly available Open Bandit Dataset (OBD) containing logged bandit propensities. Our results demonstrate that PTE-GT significantly outperforms strong baselines. Critically, on datasets with randomized exposure or logged propensities, our unbiased evaluation protocol, using SNIPS-weighted metrics, reveals a substantial performance leap that is masked by standard, biased metrics. Our method also shows marked improvements in model calibration and long-tail item recommendation.

Keywords: graph transformer; session-based recommendation; debiasing; GNNS; inverse propensity scoring; contrastive learning; randomized exposure; digital economy

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1. Introduction

Session-based recommender (SBR) systems are a critical component of modern online platforms, including e-commerce, video streaming, and content delivery [1–3]. Their

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Article 数字经济中基于会话的去偏推荐：图转换器的倾向感知训练和时间对比

Yongjian Wang [1], Junru Si [2], Xuhua Qiu [2,*] and Kunjie Zhu [2,3,*]

[1] School of Foreign Languages, Guangdong Polytechnic Normal University, Guangzhou 510640, China; wangyongjian@gpnu.edu.cn

[2] Institutes of Science and Development, Chinese Academy of Sciences, Beijing 100190, China; sijunru25@mails.ucas.ac.cn

[3] Department of Mathematics, City University of Hong Kong, Hong Kong, China

* Correspondence: qiuxuhua23@mails.ucas.ac.cn (X.Q.); kunjie.zhu@my.cityu.edu.hk (K.Z.)

Abstract

primary goal is to predict a user's next action, such as the next item to click or purchase, based solely on the sequence of interactions within their current, often anonymous, session. Unlike traditional collaborative filtering, SBR models must operate without access to long-term user profiles or explicit user identifiers, making them essential for handling new users and respecting user privacy [4].

To capture the complex dependencies within these short-term interaction sequences, the field has increasingly shifted towards powerful deep learning architectures. Early successes with recurrent neural networks (RNNs) were followed by the adoption of Transformer models, such as the Self-Attentive Sequential Recommendation (SASRec) architecture [5], which leverages self-attention to capture long-range dependencies between items in a sequence. Concurrently, Graph Neural Networks (GNNs) have demonstrated significant efficacy. This paradigm is motivated by the observation that user behavior is often not strictly linear. GNNs address this by modeling a session as an item-transition graph [6–8]. This graph-based approach, exemplified by the Session-based Recommendation Graph Neural Network (SR-GNN), allows the model to explicitly capture complex, non-sequential relationships and structural patterns, such as revisiting items or exploring divergent paths. By propagating information along the graph's edges, GNNs can model dependencies between items that are distant in time but semantically related, a task that remains challenging for purely sequential models. Recent work has sought to integrate these paradigms, resulting in advanced backbones like graph transformers that leverage GNN message passing within an attention-based architecture, jointly modeling sequential context and structural connectivity [9]. Subsequent works further refined these ideas by incorporating target-aware attention mechanisms [10], dual-channel hypergraph modeling [11], and lossless encoding schemes [12].

Despite their modeling sophistication, these systems are almost universally trained on observational data that suffers from a fundamental flaw: exposure bias [13–15]. The interaction logs collected from production systems do not represent a uniform sample of true user preferences. Instead, they reflect a heavily biased feedback loop, where items displayed at higher ranks or with greater frequency are clicked more often, simply because they are seen more. This bias is inherent in the logging policy. Models trained naively on such data learn to conflate this visibility-driven popularity with genuine user relevance [16,17]. Consequently, these models tend to overfit to the biases of the logging policy, leading to sub-optimal recommendations, poor generalization to new or long-tail items, and a significant distortion of offline evaluation metrics.

In the broader field of Learning-to-Rank (LTR), mitigating such biases through counterfactual learning is a well-established practice [18]. Methods based on inverse propensity scoring (IPS) can, under certain assumptions, correct for the biased data distribution by re-weighting the training loss for each sample [19]. This allows the model to optimize for an unbiased risk estimate. However, the application of these principled debiasing techniques in the session-based recommendation domain has been severely limited. This is particularly true for complex Graph Neural Network and Transformer models, where the non-trivial architectures and the graph-based data representation add complexity to the integration of sample-wise propensity scores.

The emergence of public datasets containing logs from randomized controlled trials, such as KuaiRand, provides a unique pathway to address this challenge [20,21]. By intentionally inserting uniformly random items into recommendation feeds, these datasets make it possible to estimate exposure propensities from data, thereby enabling the use of counterfactual estimation techniques.

In this work, we leverage this capability to develop a principled framework for debiasing session-based graph recommendation. We adopt a recent Interval-enhanced

主要目标是仅根据用户当前（通常是匿名）会话中的交互顺序来预测用户的下一步操作，例如点击或购买的下一个商品。与传统的协同过滤不同，SBR 模型必须在无法访问长期用户配置文件或显式用户标识符的情况下运行，这使得它们对于处理新用户和尊重用户隐私至关重要 [4]。为了捕获这些短期交互序列中的复杂依赖关系，该领域越来越多地转向强大的深度学习架构。循环神经网络（RNN）的早期成功之后是 Transformer 模型的采用，例如自注意力序列推荐（SASRec）架构[5]，它利用自注意力来捕获序列中项目之间的远程依赖关系。同时，图神经网络（GNN）已表现出显着的功效。这种范例的动机是观察到用户行为通常不是严格线性的。 GNN 通过将会话建模为项目转换图来解决这个问题 [6-8]。这种基于图的方法，以基于会话的推荐图神经网络（SR-GNN）为例，允许模型显式捕获复杂的、非顺序的关系和结构模式，例如重新访问项目或探索不同的路径。通过沿着图的边缘传播信息，GNN 可以对时间上遥远但语义相关的项目之间的依赖关系进行建模，这对于纯顺序模型来说仍然具有挑战性。最近的工作试图整合这些范式，从而产生像图转换器这样的高级主干，它们利用基于注意力的架构内的 GNN 消息传递，联合建模顺序上下文和结构连接[9]。随后的工作通过结合目标感知注意机制[10]、双通道超图建模[11]和无损编码方案[12]进一步完善了这些想法。尽管建模非常复杂，但这些系统几乎普遍接受了观测数据的训练，但存在一个根本缺陷：曝光偏差[13-15]。从生产系统收集的交互日志并不代表真实用户偏好的统一样本。相反，它们反映了一个严重偏见的反馈循环，其中排名较高或频率较高的项目被更频繁地点击，仅仅是因为它们被看到的次数更多。这种偏见是伐木政策所固有的。根据此类数据进行天真训练的模型学会了将这种可见性驱动的流行度与真正的用户相关性混为一谈[16,17]。因此，这些模型往往会过度适应日志记录策略的偏差，从而导致次优推荐、对新项目或长尾项目的泛化能力较差以及离线评估指标的严重扭曲。在更广泛的排名学习（LTR）领域，通过反事实学习减轻这种偏见是一种行之有效的做法[18]。基于逆倾向评分（IPS）的方法可以在某些假设下通过重新加权每个样本的训练损失来纠正有偏差的数据分布[19]。这使得模型能够针对无偏风险估计进行优化。然而，这些有原则的去偏差技术在基于会话的推荐领域的应用受到了严重限制。对于复杂的图神经网络和 Transformer 模型来说尤其如此，其中不平凡的架构和基于图的数据表示增加了样本倾向得分集成的复杂性。包含随机对照试验日志的公共数据集（例如 KuaiRand）的出现，为解决这一挑战提供了独特的途径 [20,21]。通过有意地将均匀随机的项目插入推荐源中，这些数据集使得可以从数据中估计暴露倾向，从而使反事实估计技术的使用成为可能。在这项工作中，我们利用这种功能来开发一个原则框架，以消除基于会话的图形推荐的偏差。我们采用了最近的 Interval-enhanced https://doi.org/10.3390/ electronics15010084

Graph Transformer as our backbone. This choice is deliberate, as the graph transformer architecture is uniquely suited to SBR by inheriting the structural reasoning power of GNNs while also using self-attention to capture long-range, time-aware dependencies. It is chosen for its ability to effectively model both the graph structure and the precise temporal intervals between events in a session. We integrate this backbone with two key training-time modules. First, we introduce a propensity-aware optimization objective based on the self-normalized inverse propensity scoring estimator, SNIPS, which uses propensities learned from randomized exposure data to correct for bias. Second, to further improve the quality and robustness of the learned session representations, we propose a lightweight, view-free temporal consistency contrastive regularizer. This module encourages the model to learn invariant representations for temporally adjacent sub-sessions (prefixes and suffixes) without relying on computationally expensive graph augmentations, which are a common bottleneck for GNN-based contrastive learning.

Our contributions are three-fold:

- We propose a principled, propensity-aware training objective grounded in unbiased counterfactual risk minimization, adapting the self-normalized inverse propensity scoring estimator for the training of an interval-aware graph transformer for session-based recommendation.
- We design a view-free temporal consistency contrastive objective that regularizes session representations by enforcing alignment between session prefixes and suffixes, improving robustness without the computational overhead of structural data augmentations, a significant challenge for graph-based models.
- We conduct a comprehensive evaluation on four public datasets—KuaiRand, OTTO, YOOCHOOSE, and the Open Bandit Dataset (OBD)—demonstrating the efficacy of our approach using both standard top-K metrics and unbiased evaluation protocols enabled by randomized exposure or logged behavior-policy propensities.

The remainder of this paper is structured as follows. We first review the related literature in session-based recommendation, unbiased learning, and contrastive methods. We then describe the datasets used in our evaluation, including those with randomized exposure. Following this, we detail our proposed methodology, covering the interval-aware graph transformer backbone, the propensity-aware learning objective, and the temporal consistency regularizer. Subsequently, we outline the experimental setup and present our comprehensive results, including performance on standard benchmarks, unbiased evaluations, and analyses of model calibration and long-tail performance. We then provide ablation studies to validate our design choices. Finally, we conclude the paper with a discussion of our findings.

2. Related Work

Session-based recommendation has progressed from recurrent models like GRU4Rec to more complex graph-based and Transformer-based architectures [22]. GNNs for SBR, such as the canonical SR-GNN, model sessions as item-transition graphs, capturing structural patterns [23]. Subsequent works further refined this approach by incorporating target-aware attention mechanisms [10], dual-channel hypergraph modeling [11], and lossless encoding schemes [12].

Concurrently, Transformer models became powerful non-graph alternatives. SASRec applied uni-directional self-attention, while Bidirectional Encoder Representations from Transformers for Recommendation (BERT4Rec) used bidirectional masking. Industrial-scale models like TRON (the production Transformer baseline released for the OTTO challenge) demonstrated the efficiency and accuracy of optimized Transformers on large datasets like OTTO, establishing strong, reproducible baselines [24]. Recently, graph transformers, such

 Graph Transformer 作为我们的骨干。这种选择是经过深思熟虑的，因为图转换器架构继承了GNN 的结构推理能力，同时还使用自注意力来捕获远程、时间感知的依赖关系，因此非常适合 SBR。选择它是因为它能够有效地对图结构和会话中事件之间的精确时间间隔进行建模。我们将此骨干网与两个关键的训练时间模块集成在一起。首先，我们引入了基于自我归一化逆倾向评分估计器 SNIPS 的倾向感知优化目标，该目标使用从随机暴露数据中学习的倾向来纠正偏差。其次，为了进一步提高学习会话表示的质量和鲁棒性，我们提出了一种轻量级、无视图时间一致性对比正则化器。该模块鼓励模型学习时间相邻子会话（前缀和后缀）的不变表示，而不依赖于计算成本高昂的图增强，这是基于 GNN 的对比学习的常见瓶颈。我们的贡献有三方面：

- 我们提出了一个基于无偏反事实风险最小化的有原则的倾向感知训练目标，采用自我归一化逆倾向评分估计器来训练区间感知图转换器以进行基于会话的推荐。

- 我们设计了一个无视图时间一致性对比目标，通过强制会话前缀和后缀之间的对齐来规范会话表示，提高鲁棒性，而无需结构数据增强的计算开销，这对基于图的模型来说是一个重大挑战。

- 我们对四个公共数据集（KuaiRand、OTTO、YOOCHOOSE 和 Open Bandit 数据集 (OBD)）进行了全面评估，使用标准 top-K 指标和通过随机暴露或记录的行为策略倾向启用的无偏见评估协议来证明我们方法的有效性。本文的其余部分结构如下。我们首先回顾了基于会话的推荐、无偏学习和对比方法的相关文献。然后，我们描述评估中使用的数据集，包括随机暴露的数据集。接下来，我们详细介绍了我们提出的方法，涵盖区间感知图转换器主干、倾向感知学习目标和时间一致性正则化器。随后，我们概述了实验设置并展示了我们的综合结果，包括标准基准的性能、公正的评估以及模型校准和长尾性能的分析。然后我们提供消融研究来验证我们的设计选择。最后，我们通过讨论我们的发现来结束本文。 2.相关工作基于会话的推荐已经从GRU4Rec等循环模型发展到更复杂的基于图和基于Transformer的架构[22]。 SBR 的 GNN，例如规范的 SR-GNN，将会话建模为项目转换图，捕获结构模式 [23]。随后的工作通过结合目标感知注意机制[10]、双通道超图建模[11]和无损编码方案[12]进一步完善了这种方法。同时，Transformer 模型成为强大的非图替代品。 SASRec 应用单向自注意力，而来自 Transformers forRecommendation 的双向编码器表示 (BERT4Rec) 使用双向掩蔽。 TRON（为 OTTO 挑战赛发布的生产 Transformer 基线）等工业规模模型展示了优化 Transformer 在 OTTO 等大型数据集上的效率和准确性，建立了强大的、可重复的基线 [24]。最近，图形转换器，例如 https://doi.org/10.3390/electronics15010084

as the Interval-enhanced Graph Transformer (IGT), have emerged to integrate structural graph information with temporal dynamics by encoding inter-event time intervals directly into the attention mechanism, aligning well with the time-aware nature of SBR [25].

A primary challenge in SBR is the inherent exposure bias in logged implicit feedback. The field of unbiased LTR provides a principled foundation for addressing this, primarily through IPS and its self-normalized variant SNIPS [26,27]. These methods re-weight training samples to correct for biased exposure, enabling the optimization of an unbiased risk estimate. Adapting these counterfactual learning techniques to recommender systems, particularly for debiasing from missing-not-at-random feedback, has become an active area of research [28].

A critical enabler for such debiasing is access to data with randomized exposure or logged behavior-policy propensities. The KuaiRand dataset, by inserting uniformly random items into feeds, provides the necessary ground truth for estimating propensities and performing reliable, SNIPS-based unbiased offline evaluation. Complementary to this, the Open Bandit Dataset (OBD) offers real-world logged bandit feedback with known action probabilities ("propensity_score") under multiple behavior policies, enabling direct off-policy evaluation and unbiased weighting [29,30]. This complements large-scale observational datasets like OTTO, which serves as a standard benchmark for session model performance under biased, real-world traffic [20,21].

Finally, contrastive learning (CL) has been widely adopted as a regularization technique. Early methods like Self-supervised Graph Learning (SGL) applied graph augmentations to create different views for contrast [31]. However, recent work such as Simple Graph Contrastive Learning (SimGCL) has questioned the necessity of complex structural augmentations, showing that simple embedding-space noise can be sufficient [32]. This motivates "view-free" approaches, which avoid the computational cost and potential artifacts of heavy augmentations. In parallel, consistency-training objectives that enforce agreement between multiple stochastic views of the same sequence have been shown to be effective for sequential recommendation without explicit data augmentations [33]. For discrete event sequences, CoLES [34] formalizes a prefix–suffix style subsequence cropping scheme and proves that, under mild assumptions, such augmentations preserve the underlying latent generative process, providing a theoretical footing for sequence-level contrastive and consistency training.

Beyond classical IPS and SNIPS, a growing line of work applies causal inference and doubly robust techniques to recommendation. CausalRec [35] introduces a front-door style visual causal adjustment to mitigate presentation bias in visually aware recommendation. Doubly robust (DR) and multiple robust (MR) estimators [36,37] combine IPS with reward modeling to reduce variance and improve robustness to propensity misspecification. Recent surveys on causal recommendation [38,39] further systematize potential-outcome-based, SCM-based, and general counterfactual approaches. Our PTE-GT is orthogonal to these architectures: it can be viewed as a plug-in, SNIPS-based training objective for an interval-aware graph transformer backbone. In Section 6.2, we empirically compare PTE-GT against IPS-enhanced SASRec, a CausalRec-style baseline, and a doubly robust model, and find that PTE-GT achieves consistently higher SNIPS-weighted accuracy on both KuaiRand and OBD.

Our work synthesizes these three advancements. We utilize the interval-aware graph transformer IGT as our backbone, apply the SNIPS counterfactual training objective enabled by randomized exposure data, and introduce a lightweight, view-free temporal consistency regularizer tailored for session graphs.

 作为间隔增强图变换器（IGT），通过将事件间时间间隔直接编码到注意机制中，将结构图信息与时间动态集成起来，与 SBR 的时间感知性质很好地结合起来[25]。 SBR 的主要挑战是记录的隐式反馈中固有的暴露偏差。无偏 LTR 领域主要通过 IPS 及其自我归一化变体 SNIPS 为解决此问题提供了原则基础[26,27]。这些方法重新加权训练样本以纠正有偏差的暴露，从而实现无偏差风险估计的优化。将这些反事实学习技术应用于推荐系统，特别是消除非随机缺失反馈的偏差，已成为一个活跃的研究领域[28]。这种消除偏差的关键推动因素是访问具有随机暴露或记录的行为政策倾向的数据。 KuaiRand 数据集通过将均匀随机项插入到 feed 中，为估计倾向和执行可靠的、基于 SNIPS 的无偏离线评估提供了必要的基本事实。与此互补的是，开放强盗数据集（OBD）提供真实世界记录的强盗反馈以及多种行为策略下的已知动作概率（“propensity_score”），从而实现直接的离策略评估和无偏权重[29,30]。这补充了 OTTO 等大规模观测数据集，OTTO 可以作为有偏差的真实流量下会话模型性能的标准基准 [20,21]。最后，对比学习（CL）作为一种正则化技术被广泛采用。自监督图学习（SGL）等早期方法应用图增强来创建不同的对比视图[31]。然而，简单图对比学习（SimGCL）等最近的工作对复杂结构增强的必要性提出了质疑，表明简单的嵌入空间噪声就足够了[32]。这激发了“无视图”方法，避免了大量增强的计算成本和潜在的伪影。同时，一致性训练目标可以强制同一序列的多个随机视图之间达成一致，这已被证明对于无需显式数据增强的顺序推荐是有效的[33]。对于离散事件序列，CoLES [34] 形式化了前缀-后缀样式的子序列裁剪方案，并证明，在温和的假设下，这种增强保留了潜在的潜在生成过程，为序列级对比和一致性训练提供了理论基础。除了经典的 IPS 和 SNIPS 之外，越来越多的工作将因果推理和双重鲁棒技术应用于推荐。CausalRec [35] 引入了前门式视觉因果调整，以减轻视觉感知推荐中的呈现偏差。双稳健 (DR) 和多重稳健 (MR) 估计器 [36,37] 将 IPS 与奖励建模相结合，以减少方差并提高对倾向错误指定的稳健性。最近关于因果推荐的调查 [38,39] 进一步系统化了基于潜在结果、基于 SCM 和一般反事实方法。我们的 PTE-GT 与这些架构是正交的：它可以被视为一个插件、基于 SNIPS 的训练目标，用于区间感知图转换器主干。在第 6.2 节中，我们根据经验将 PTE-GT 与 IPS 增强型 SASRec（CausalRec 式基线和双鲁棒模型）进行比较，发现 PTE-GT 在 KuaiRand 和 OBD 上始终实现更高的 SNIPS 加权精度。我们的工作综合了这三项进步。我们利用间隔感知图转换器 IGT 作为主干，应用由随机暴露数据支持的 SNIPS 反事实训练目标，并引入专为会话图定制的轻量级、无视图时间一致性正则化器。 https://doi.org/10.3390/ electronics15010084

3. Datasets

We evaluate on three widely used public session-based datasets (KuaiRand, OTTO, and YOOCHOOSE-1/64) and additionally on the publicly available Open Bandit Dataset (OBD), which provides logged bandit feedback with known propensities. KuaiRand provides intervened logs with randomized exposure suitable for unbiased training and evaluation; OTTO offers industry-grade, multi-objective sessions at scale; and YOOCHOOSE-1/64 remains a canonical session benchmark with widely reported statistics. OBD supplies impression-level logs with behavior-policy propensity scores suitable for off-policy evaluation and debiased training. Table 1 summarizes the descriptive statistics. KuaiRand's statistics are reported at the user-sequence level, as they are released as continuous user histories rather than pre-segmented sessions.

Table 1. Dataset descriptive statistics. Note: OBD consists of impression-level bandit logs with logged propensities; session-length metrics are not directly applicable.

Dataset (Split)	Sequence Unit	Sequences	Items	Interactions	Avg. Length	Randomized Propensities
KuaiRand-27K	user history	27,285	32,038,725	322,278,385	—	Yes
Open Bandit Dataset	impression log	26 M	80	—	—	Yes (logged bandit)
OTTO (train)	session	12,899,779	1,855,603	216,716,096	16.80	No
YOOCHOOSE-1/64	session	369,859	16,766	557,248	6.16	No

KuaiRand aggregates rich feedback signals from a large video platform and, crucially, inserts random items into normal feeds so that a small share of impressions is assigned uniformly at random. The official documentation reports three releases; we use KuaiRand-27K as our default and draw unbiased evaluation slices from logs flagged as randomized. The site specifies that approximately 0.37% of interactions in the final data are replaced by randomly sampled items, and it lists per-release counts for users, items, and interactions that we adopt here. These properties make KuaiRand suitable for propensity estimation and SNIPS-weighted evaluation.

OTTO is a real-world e-commerce session dataset collected from an online retailer and published with a Kaggle task. The maintainers provide aggregate counts for the training split: 12.9 million sessions, 1.86 million unique items, and 216.7 million events, with the event mix broken down into clicks, cart additions, and orders, and with descriptive distributional statistics such as a mean of 16.80 events per session. We follow the single-objective next-click protocol when aligning with session-based literature.

YOOCHOOSE-1/64 is the most common compact split of the RecSys-2015 challenge dataset used for session-based recommendation. Following the authoritative SR-GNN study, we report 369,859 training sessions and 55,898 test sessions, with 16,766 unique items, 557,248 click events, and an average session length of 6.16. We preserve the standard preprocessing that removes length-1 sessions and constructs next-item labels via prefix expansion.

Open Bandit Dataset (OBD). We additionally use the Open Bandit Dataset, a large-scale logged bandit dataset collected from the ZOZOTOWN fashion e-commerce platform. OBD provides impression-level logs with multiple behavior policies and records the action probabilities as a "propensity_score" field, along with click indicators. We use the "all" campaign and follow the official release guidance for preprocessing; this dataset is well suited for off-policy evaluation and propensity-aware training [29,40]. In our preprocessed split, the logged action set size $|\mathcal{A}_i|$ is almost always three: 25.48 M impressions (97.99%) have exactly three candidate actions, 0.41 M (1.58%) have two, and only 0.11 M (0.42%) have a single action due to logging artefacts. To adapt this impression-level bandit data to our session-style encoder, we sort impressions chronologically for each user and construct

 3. 数据集 我们对三个广泛使用的基于会话的公共数据集（KuaiRand、OTTO 和 YOOCHOOSE-1/64）以及公开可用的开放强盗数据集 (OBD) 进行评估，该数据集提供了记录的具有已知倾向的强盗反馈。快兰德提供随机暴露的干预日志，适合公正的训练和评估； OTTO 提供大规模的行业级、多目标会议； YOOCHOOSE-1/64 仍然是具有广泛报告统计数据的规范会话基准。 OBD 提供印象级日志，其中包含适合偏离政策评估和去偏训练的行为政策倾向评分。表 1 总结了描述性统计数据。 KuaiRand 的统计数据是在用户序列级别报告的，因为它们是作为连续的用户历史而不是预先分段的会话发布的。

Table 1. Dataset descriptive statistics. Note: OBD consists of impression-level bandit logs with logged propensities; session-length metrics are not directly applicable.

Dataset (Split)	Sequence Unit	Sequences	Items	Interactions	Avg. Length	Randomized Propensities
KuaiRand-27K	user history	27,285	32,038,725	322,278,385	—	Yes
Open Bandit Dataset	impression log	26 M	80	—	—	Yes (logged bandit)
OTTO (train)	session	12,899,779	1,855,603	216,716,096	16.80	No
YOOCHOOSE-1/64	session	369,859	16,766	557,248	6.16	No

KuaiRand聚合了来自大型视频平台的丰富反馈信号，最重要的是，将随机项目插入到正常的Feed中，以便随机均匀分配一小部分展示次数。官方文档报告了三个版本；我们使用 KuaiRand-27K 作为默认值，并从标记为随机的日志中绘制无偏评估切片。该网站指定最终数据中大约 0.37% 的交互被随机采样的项目替换，并列出了我们在此处采用的用户、项目和交互的每次发布计数。这些特性使得 KuaiRand 适合倾向估计和 SNIPS 加权评估。 OTTO 是从在线零售商收集并通过 Kaggle 任务发布的真实电子商务会话数据集。维护者提供了训练分割的总计计数：1290 万个会话、186 万个独特项目和 2.167 亿个事件，事件组合细分为点击、购物车添加和订单，以及描述性分布统计数据，例如每个会话平均 16.80 个事件。在与基于会话的文献保持一致时，我们遵循单目标下次点击协议。 YOOCHOOSE-1/64 是用于基于会话的推荐的 RecSys-2015 挑战数据集最常见的紧凑分割。根据权威的 SR-GNN 研究，我们报告了 369,859 个训练会话和 55,898 个测试会话，其中有 16,766 个独特项目、557,248 个点击事件，平均会话长度为 6.16。我们保留标准预处理，删除长度为 1 的会话并通过前缀扩展构造下一项标签。打开 Bandit 数据集 (OBD)。我们还使用开放强盗数据集，这是从 ZOZOTOWN 时尚电子商务平台收集的大规模记录强盗数据集。 OBD 提供具有多种行为策略的印象级日志，并将操作概率记录为“propensity_score”字段以及点击指示器。我们使用“all”竞选并遵循官方发布指南进行预处理；该数据集非常适合离策略评估和倾向感知训练[29,40]。在我们的预处理分割中，记录的操作集大小 $|\mathcal{A}_i|$ 几乎总是 3：2548 万次展示 (97.99%) 恰好有 3 个候选操作，0.41 M (1.58%) 有两个候选操作，由于记录伪影，只有 0.11 M (0.42%) 有一个操作。为了使这种印象级强盗数据适应我们的会话式编码器，我们按时间顺序对每个用户的印象进行排序并构建 https://doi.org/10.3390/electronics15010084

a sliding context of the last w impressions. The session encoder f_θ receives the sequence of the last w (item, timestamp) pairs and outputs a context embedding, which is then scored against the three candidate actions in $\mathcal{A}_i$. We choose a window size of $w = 10$ based on validation performance on OBD, as discussed in Section 7.3. For each impression i with logged action set $\mathcal{A}_i$ and click label, we score all $a \in \mathcal{A}_i$ using the same encoder fed with the last w impressions as context. Top-K is computed over $\mathcal{A}_i$; the clicked action (if any) serves as the ground truth. SNIPS weights $\frac{1}{\texttt{propensity_score}_i}$ are applied to the indicator terms in Recall@K and to the reciprocal rank in MRR@K, with self-normalization across the evaluation set.

4. Methodology

This section presents a principled framework for debiasing session-based recommendation with a recent graph-transformer backbone and two training-time modules. The pipeline consists of (i) constructing a temporally enriched session graph and encoding it with an interval-aware graph transformer; (ii) learning under inverse-propensity weighting estimated from logs with randomized exposure; and (iii) regularizing the session representation by a view-free temporal consistency objective. Figure 1 illustrates the end-to-end framework, Figure 2 details the propensity-aware learning flow, and Figure 3 depicts the temporal consistency mechanism.

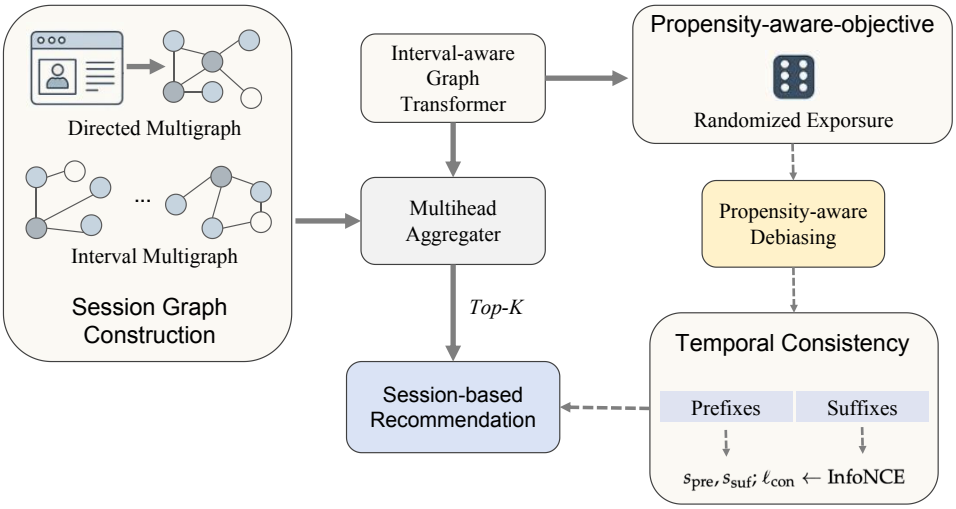

Figure 1. The end-to-end framework illustrating session graph construction, the interval-aware graph transformer, propensity-aware objective, and temporal consistency module.

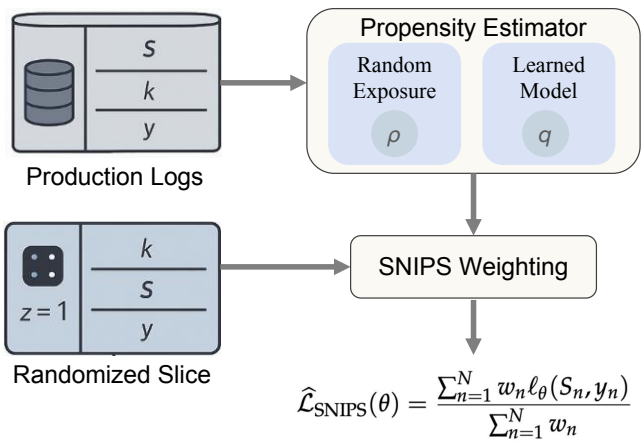

Figure 2. The propensity-aware learning flow, showing propensity estimation from randomized logs and the application of SNIPS weighting.

最后 w 展示次数的滑动上下文。会话编码器 f_θ 接收最后一个 w (项的序列、时间戳) 对并输出上下文嵌入，然后根据 $\mathcal{A}_i$ 中的三个候选动作对其进行评分。我们根据 OBD 上的验证性能选择 $w = 10$ 的窗口大小，如第 7.3 节所述。对于具有记录的操作集 $\mathcal{A}_i$ 和点击标签的每个展示 i，我们使用以最后 w 展示作为上下文的同一编码器对所有 $a \in \mathcal{A}_i$ 进行评分。 Top-K 是在 $\mathcal{A}_i$ 上计算的；单击的操作（如果有）作为基本事实。 SNIPS 权重 ^{1} propensity_score$_i$ 应用于 Recall@K 中的指示项以及 MRR@K 中的倒数排名，并在整个评估集中进行自我归一化。 4.方法本节介绍了一个原则框架，用于消除基于会话的推荐的偏差，其中包括最近的图形转换器主干和两个训练时间模块。该管道包括（i）构建一个时间丰富的会话图并使用间隔感知图转换器对其进行编码； (ii) 根据随机暴露的日志估计的逆倾向权重下的学习； (iii) 通过无视图时间一致性目标来规范会话表示。图 1 说明了端到端框架，图 2 详细介绍了倾向感知学习流程，图 3 描述了时间一致性机制。

Figure 1. The end-to-end framework illustrating session graph construction, the interval-aware graph transformer, propensity-aware objective, and temporal consistency module.

Figure 2. The propensity-aware learning flow, showing propensity estimation from randomized logs and the application of SNIPS weighting.

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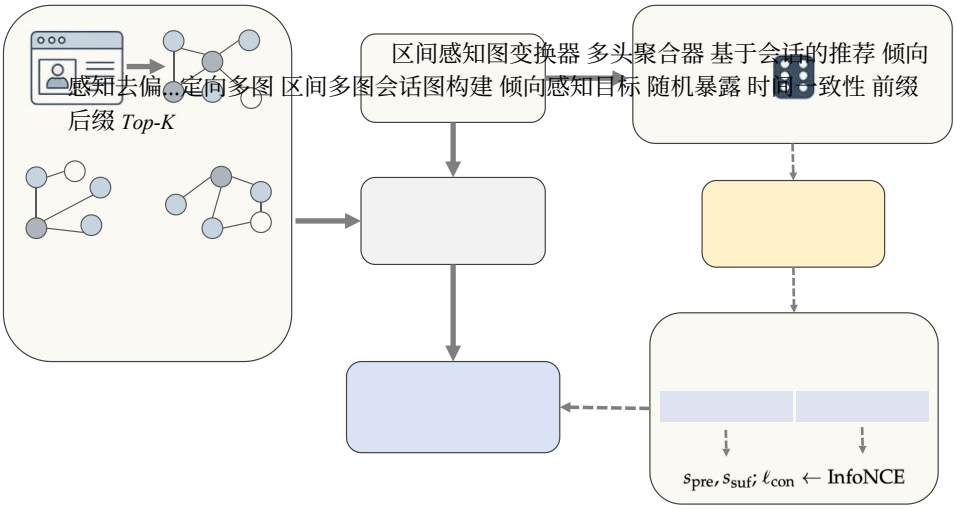
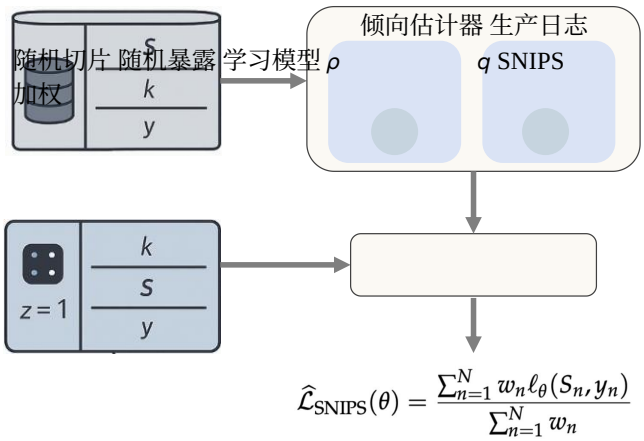




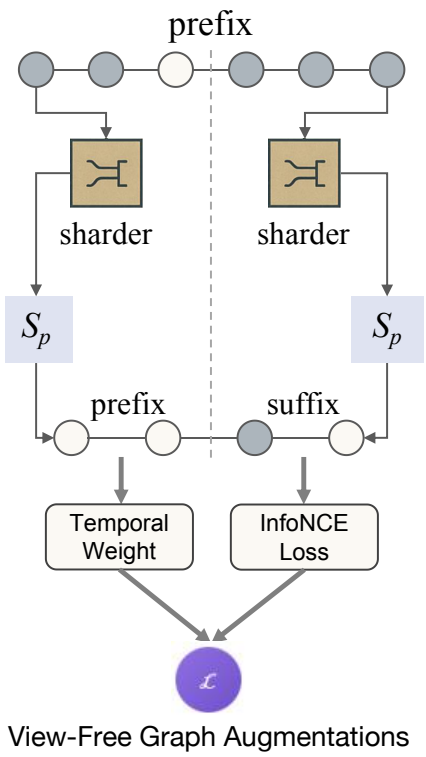


Figure 3. The view-free temporal consistency mechanism, showing the prefix–suffix split, temporal decay weighting, and contrastive alignment.

4.1. Session Graph Construction and Interval-Aware Graph Transformer

Our session encoder is based on the Interval-enhanced Graph Transformer (IGT) architecture [25]. Concretely, we adopt the same interval-aware multi-head attention mechanism and Fourier time encoding as in IGT but restrict the neighborhood $\mathcal{N}(i)$ to one-step predecessors and successors in the session graph and use gated time-aware readout in place of the original global pooling.

Let a user session be an ordered sequence $S = (x_1, \ldots, x_T)$ of item interactions with timestamps $(t_1, \ldots, t_T)$. We construct a directed multigraph $G = (V, E)$ per session, where $V = \{x_1, \ldots, x_T\}$ and $E = \{(x_\ell \to x_{\ell+1})\}_{\ell=1}^{T-1}$. Each edge carries a time interval feature $\Delta t_\ell = t_{\ell+1} - t_\ell$ and a relative rank feature $\Delta p_\ell = 1$. Node features are $h_\ell^{(0)} = e(x_\ell) + p(\ell)$, where $e(\cdot)$ is the learnable item embedding and $p(\ell)$ is a learnable position embedding within the session.

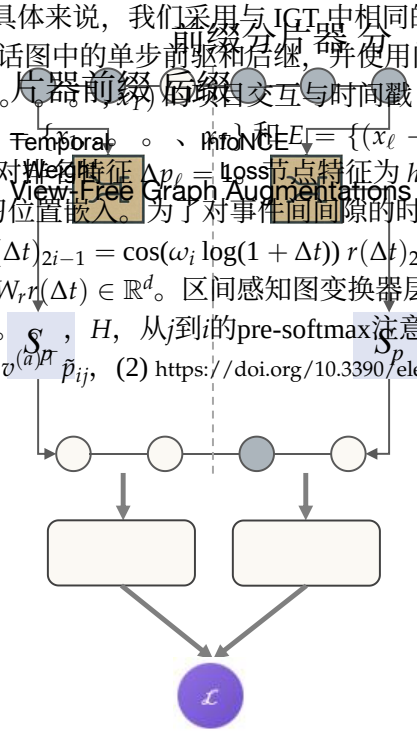
To encode temporal recency and nonlinearity of inter-event gaps, we map Δt to a Fourier time feature

$$\begin{aligned} r(\Delta t)_{2i-1} &= \cos(\omega_i \log(1 + \Delta t)) \\ r(\Delta t)_{2i} &= \sin(\omega_i \log(1 + \Delta t)) \end{aligned} \quad \text{for } i = 1, \ldots, m \tag{1}$$

followed by a linear projection $\tilde{r} = W_r r(\Delta t) \in \mathbb{R}^d$. The interval-aware graph transformer layer updates node representations with multi-head attention over one-step predecessors and successors. For a head $a = 1, \ldots, H$, the pre-softmax attention energy from j to i is

$$e_{ij}^{(a)} = \frac{\left(Q^{(a)} h_i\right)^\top \left(K^{(a)} h_j\right)}{\sqrt{d/H}} + u^{(a)\top} \tilde{r}_{ij} + v^{(a)\top} \tilde{p}_{ij}, \tag{2}$$

图 3.无视图时间一致性机制，显示了前缀-后缀分割、时间衰减权重和对比对齐。



4.1. Session Graph Construction and Interval-Aware Graph Transformer 我们的会话编码器基于间隔增强图变换器（IGT）架构[25]。具体来说，我们采用与 IGT 中相同的间隔感知多头注意力机制和傅立叶时间编码，但将邻域 $\mathcal{N}(i)$ 限制为会话图中的单步前驱和后继，并使用门控时间感知读出代替原始全局池化。令用户会话为有序序列 $S = (x_1, \ldots, x_T)$ 的项目交互与时间戳 $(t_1, \ldots, t_T)$。我们为每个会话构造一个有向多重图 $G = (V, E)$，其中 $V = \{x_1, \ldots, x_T\}$ 和 $E = \{(x_\ell \to x_{\ell+1})\}_{\ell=1}^{T-1}$。每条边都带有时间间隔特征 $\Delta t_\ell = t_{\ell+1} - t_\ell$ 和相对位置特征 $\Delta p_\ell = 1$。节点特征为 $h_\ell^{(0)} = e(x_\ell) + p(\ell)$，其中 $e(\cdot)$ 是可学习项目嵌入，$p(\ell)$ 是会话内可学习位置嵌入。为了对事件间间隙的时间新近度和非线性进行编码，我们将 Δt 映射到 $i = 1, \ldots, m$ 的傅立叶时间特征 $r(\Delta t)_{2i-1} = \cos(\omega_i \log(1 + \Delta t))$ $r(\Delta t)_{2i} = \sin(\omega_i \log(1 + \Delta t))$。$m$ (1) 后跟线性投影 $\tilde{r} = W_r r(\Delta t) \in \mathbb{R}^d$。区间感知图变换器层通过对单步前驱和后继的多头关注来更新节点表示。对于头部 $a = 1, \ldots, H$，从 j 到 i 的 pre-softmax 注意力能量为 $e_{ij}^{(a)} = \frac{(Q^{(a)} h_i) \cdot (K^{(a)} h_j)}{\sqrt{d}} + u^{(a)\top} \tilde{r}_{ij} + v^{(a)\top} \tilde{p}_{ij}$，(2)

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where $\tilde{r}_{ij} = W_r r(\Delta t_{ij})$, $\tilde{p}_{ij} = W_p \mathrm{onehot}(\Delta p_{ij})$, and $Q^{(a)}, K^{(a)} \in \mathbb{R}^{(d/H)\times d}$ are head-specific projections. Attention coefficients are

$$\alpha_{ij}^{(a)} = \frac{\exp(e_{ij}^{(a)})}{\sum_{k\in\mathcal{N}(i)} \exp(e_{ik}^{(a)})}, \quad z_i^{(a)} = \sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(a)} V^{(a)} h_j, \tag{3}$$

and the head outputs are concatenated and passed through a position-wise feed-forward network with residual connections and layer normalization:

$$h_i' = \mathrm{LN}\Big(h_i + \mathrm{Concat}\big(z_i^{(1)}, \ldots, z_i^{(H)}\big)\Big), \quad h_i^{(l+1)} = \mathrm{LN}\Big(h_i' + \mathrm{FFN}(h_i')\Big). \tag{4}$$

Here, $\mathrm{LN}(\cdot)$ denotes layer normalization and $\mathrm{FFN}(\cdot)$ is a position-wise feed-forward network. After L layers we obtain node states $h_i^{(L)}$. The session representation is a gated readout with time-aware weights:

$$\gamma_i = \sigma\big(w^\top h_i^{(L)} + b \log(1 + \kappa \Delta t_i^{\mathrm{now}})\big), \quad s = \sum_{i=1}^{T} \frac{\gamma_i}{\sum_k \gamma_k} h_i^{(L)}, \tag{5}$$

where $\Delta t_i^{\mathrm{now}}$ is the recency gap between item x_i and the session end. The next-item distribution is $p_\theta(y \mid S) = \mathrm{softmax}(W_o s)$.

4.2. Propensity-Aware Learning Objective

Logs collected under ranked exposure are biased. Let $D = \{(S_n, y_n, a_n)\}_{n=1}^N$ be training triples with observed next item y_n and logging action a_n indicating that y_n is displayed and clicked under the production policy. The unbiased Learning-to-Rank risk for cross-entropy loss $\ell_\theta(S, y) = -\log p_\theta(y \mid S)$ can be estimated by inverse propensity scoring:

$$\widehat{\mathcal{L}}_{\mathrm{IPS}}(\theta) = \frac{1}{N} \sum_{n=1}^{N} \frac{\mathbb{I}(a_n = 1)}{\hat{p}_n} \ell_\theta(S_n, y_n), \quad \hat{p}_n \equiv \hat{p}(a = 1 \mid S_n, y_n). \tag{6}$$

To stabilize variance we adopt the self-normalized estimator

$$\widehat{\mathcal{L}}_{\mathrm{SNIPS}}(\theta) = \frac{\sum_{n=1}^N w_n \ell_\theta(S_n, y_n)}{\sum_{n=1}^N w_n}, \qquad w_n = \min\left(c, \frac{1}{\hat{p}_n}\right), \tag{7}$$

with clipping threshold $c > 0$. We further penalize weight dispersion to reduce the impact of rare large weights:

$$\mathcal{R}_{\mathrm{var}} = \lambda \cdot \frac{1}{N} \sum_{n=1}^{N} (w_n - \bar{w})^2, \quad \bar{w} = \frac{1}{N} \sum_{n=1}^{N} w_n. \tag{8}$$

In datasets that provide randomized exposure or logged behavior-policy propensities, the estimation of $\hat{p}_n$ is straightforward. For KuaiRand, we estimate exposure probabilities from the randomized insertion protocol within the logs. For OBD, we directly use the logged behavior-policy probabilities, setting $\hat{p}_n =$ `propensity_score` as recorded in the dataset, which obviates the need for a fitted exposure model and enables principled SNIPS-weighted training and evaluation. For diagnostic analysis in Section 7, we additionally evaluate the restricted and MLP-based propensity models on OBD by replacing the logged propensities with their predictions when computing the SNIPS weights, allowing us to study the impact of model capacity on calibration, variance, and downstream debiased performance. The supervised objective is $\mathcal{J}_{\mathrm{sup}} = \widehat{\mathcal{L}}_{\mathrm{SNIPS}} + \mathcal{R}_{\mathrm{var}}$.

 其中 $\tilde{r}_{ij} = W_r r(\Delta t_{ij})$、$\tilde{p}_{ij} = W_p \mathrm{onehot}(\Delta p_{ij})$ 和 $Q^{(a)}$、$K^{(a)} \in \mathbb{R}^{(d/H)\times d}$ 是头部特定的投影。注意系数为 $\alpha_{ij}^{(a)} = \exp(e_{ij}^{(a)}) \sum_{k\in\mathcal{N}(i)} \exp(e_{ik}^{(a)})$、$z_i^{(a)} = \sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(a)} V^{(a)} h_j$, (3)，头输出被连接并通过具有残差连接和层归一化的位置前馈网络：$h_i' = \mathrm{LN}\Big(h_i + \mathrm{Concat}(z_i^{(1)}, 。。。, z_i^{(H)})\Big), h_i^{(l+1)} = \mathrm{LN}\Big(h_i' + \mathrm{FFN}(h_i')\Big)$。 (4) 这里，LN(·)表示层归一化，FFN(·)是位置式前馈网络。在 L 层之后，我们获得节点状态 $h_i^{(L)}$。会话表示是具有时间感知权重的门控读数：$\gamma_i = \sigma\big(w^\top h_i^{(L)} + b \log(1 + \kappa \Delta t_i^{\mathrm{now}})\big), s = \sum_{i=1}^{T} \frac{\gamma_i}{\sum_k \gamma_k} h_i^{(L)}$, (5) 其中 $\Delta t_i^{\mathrm{now}}$ 是项目 x_i 和会话结束之间的新近度差距。下一项分布是 $p_\theta(y \mid S) = \mathrm{softmax}(W_o s)$。

4.2. Propensity-Aware Learning Objective 根据排名曝光收集的日志存在偏差。令 $D = \{(S_n、y_n、a_n)\}_{n=1}^{N}$ 为训练三元组，其中观察到的下一项 y_n 和记录操作 a_n 指示在生产策略下显示并单击 y_n。交叉熵损失 $\ell_\theta(S、y) = -\log p_\theta(y \mid S)$ 的无偏学习排序风险可以通过逆倾向评分来估计：$\widehat{\mathcal{L}}_{\mathrm{IPS}}(\theta) = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}(a_n = 1) \frac{\ell_\theta(S_n、y_n)}{\hat{p}_n}$、$\hat{p}_n \equiv \hat{p}(a = 1 \mid S_n、y_n)$。 (6) 为了稳定方差，我们采用自归一化估计器 $\widehat{\mathcal{L}}_{\mathrm{SNIPS}}(\theta) = \frac{\sum_{n=1}^{N} w_n \ell_\theta(S_n、y_n)}{\sum_{n=1}^{N} w_n}$、$w_n = \min\left(c、\frac{1}{\hat{p}_n}\right)$、(7)，裁剪阈值 $c > 0$。我们进一步惩罚权重分散以减少罕见大权重的影响：$\mathcal{R}_{\mathrm{var}} = \lambda \cdot \frac{1}{N} \sum_{n=1}^{N} (w_n - \bar{w})^2$、$\bar{w} = \frac{1}{N} \sum_{n=1}^{N} w_n$。 (8) 在提供随机暴露或记录的行为政策倾向的数据集中，$\hat{p}_n$ 的估计很简单。对于 KuaiRand，我们根据日志中的随机插入协议估计暴露概率。对于 OBD，我们直接使用记录的行为策略概率，设置数据集中记录的 $\hat{p}_n = \mathrm{propensity_score}$，这消除了对拟合暴露模型的需要，并实现了有原则的 SNIPS 加权训练和评估。对于第 7 节中的诊断分析，我们还通过在计算 SNIPS 权重时用预测替换记录的倾向来评估 OBD 上的受限倾向和基于 MLP 的倾向模型，从而使我们能够研究模型容量对校准、方差和下游除偏性能的影响。监督目标是 $\mathcal{J}_{\mathrm{sup}} = \widehat{\mathcal{L}}_{\mathrm{SNIPS}} + \mathcal{R}_{\mathrm{var}}$。 https://doi.org/10.3390/electronics15010084

Propensity estimation from randomized exposure. On logs containing randomized insertions, a binary indicator $z_n \in \{0,1\}$ marks whether item display at position k was assigned at random with known probability ρ_k. We model the exposure propensity as a convex mixture

$$\hat{p}_n = z_n \rho_{k_n} + (1 - z_n) q_\phi(S_n, y_n, k_n), \tag{9}$$

where q_ϕ is a parametric exposure model fitted on the randomized slice by maximum likelihood. A simple and effective choice is a position-and-context model

$$q_\phi(S, y, k) = \sigma\big(\alpha_k + \beta^\top f(S) + \gamma^\top g(y)\big), \tag{10}$$

with σ as the logistic link, $f(S)$ session covariates such as length, dwell-time statistics and recency histogram, and $g(y)$ item covariates such as global popularity and category. When transferring to datasets without randomized insertions, q_ϕ is applied directly, and calibration is performed by matching the empirical display rate per rank.

In practice, we instantiate three propensity models of increasing capacity on the randomized slices of KuaiRand and OBD: (i) a restricted logistic regression (LR-Restrict) that uses only rank, log global popularity, and a small set of campaign indicators as features; (ii) a full logistic regression (LR-Full) that augments these with richer session-level covariates $f(S)$ (session length, log-recency of the last click, mean and variance of dwell times over the last three interactions, and a coarse histogram of inter-arrival times) and item-level covariates $g(y)$ (log global exposure count, log click count, and a learned projection of the item's categorical ID); and (iii) a two-layer MLP (MLP-2layer) with 64 hidden units that takes the same feature vector as input. All three models are trained by maximum likelihood on randomized impressions, and their calibrated propensities are then plugged into the SNIPS objective to train PTE-GT on KuaiRand and OBD.

Unbiased evaluation. For a held-out set with randomized insertions, we compute SNIPS-weighted Recall@K and MRR@K by replacing each indicator with its IPS-weighted counterpart, ensuring that offline evaluation does not inflate scores due to rank bias. Figure 2 summarizes the estimation and weighting flow. Propensity model comparisons are summarized in Table 2.

Table 2. Comparison of propensity models on KuaiRand and OBD. All SNIPS-R@20 and Var(w) values use clipping threshold $c = 20$.

Dataset	Propensity Model	AUC	ECE (%)	Brier (%)	SNIPS-R@20 (%)	Var(w)
KuaiRand	LR-Restrict	0.74	2.80	7.50	12.47	0.83
KuaiRand	LR-Full (ours)	0.78	1.90	7.12	13.05	1.00
KuaiRand	MLP-2layer	0.80	1.60	7.05	13.09	1.90
OBD	LR-Restrict	0.76	2.50	6.50	12.00	0.78
OBD	LR-Full (ours)	0.80	1.70	6.20	12.74	1.00
OBD	MLP-2layer	0.82	1.50	6.15	12.81	1.85

4.3. Temporal Consistency Contrast Without Heavy Augmentations

To improve robustness and long-tail generalization, we impose a view-free consistency constraint on session representations. For each session S we form a prefix $S^{\text{pre}} = (x_1, \ldots, x_\tau)$ and a suffix $S^{\text{suf}} = (x_{\tau+1}, \ldots, x_T)$ with $\tau \sim \text{Unif}\{2, \ldots, T-2\}$. Passing each through the same encoder yields embeddings $s^{\text{pre}}, s^{\text{suf}} \in \mathbb{R}^d$. We add a small latent perturbation $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ to both views to avoid collapse while avoiding structural graph augmentations. Intuitively, this loss encourages the encoder to map different contiguous subsequences (prefix and suffix) of the same session to nearby points in the representation space,

 根据随机暴露进行倾向估计。在包含随机插入的日志上，二进制指示符 $z_n \in \{0, 1\}$ 标记位置 k 处的项目显示是否以已知概率 ρ_k 随机分配。我们将暴露倾向建模为凸混合 $\hat{p}_n = z_n \rho_{k_n} + (1 - z_n) q_\phi(S_n, y_n, k_n)$，(9)，其中 q_ϕ 是通过最大似然拟合在随机切片上的参数暴露模型。一个简单而有效的选择是位置和上下文模型 $q_\phi(S、y、k) = \sigma\big(\alpha_k + \beta^\top f(S) + \gamma^\top g(y)\big)$、(10)，其中 σ 作为逻辑链接，$f(S)$ 会话协变量（例如长度、停留时间统计数据和新近度直方图）以及 $g(y)$ 项目协变量（例如全球流行度和类别）。当转移到没有随机插入的数据集时，直接应用q_ϕ，并通过匹配每个等级的经验显示率来执行校准。在实践中，我们在 KuaiRand 和 OBD 的随机切片上实例化了三种增加容量的倾向模型：（i）仅使用排名、日志全球流行度和一小组活动指标作为特征的受限逻辑回归（LR-Restrict）； (ii) 完整的逻辑回归 (LR-Full)，通过更丰富的会话级协变量$f(S)$ (会话长度、最后一次点击的对数新近度、过去 3 次交互的停留时间的平均值和方差、到达间隔时间的粗略直方图)和项目级协变量$g(y)$ (日志全局曝光计数、日志点击计数以及项目分类 ID 的学习投影)来增强这些变量； (iii) 具有 64 个隐藏单元的两层 MLP (MLP-2layer)，采用相同的特征向量作为输入。所有三个模型都通过随机印象的最大似然进行训练，然后将它们的校准倾向插入 SNIPS 目标中，以在 KuaiRand 和 OBD 上训练 PTE-GT。公正的评价。对于具有随机插入的保留集，我们通过用 IPS 加权对应指标替换每个指标来计算 SNIPS 加权 Recall@K 和 MRR@K，确保离线评估不会因排名偏差而夸大分数。图 2 总结了估计和加权流程。表2总结了倾向模型的比较。 表2. KuaiRand和OBD倾向模型的比较。所有 SNIPS-R@20 和 Var(w) 值均使用裁剪阈值 $c = 20$。数据集倾向模型 AUC ECE (%) Brier (%) SNIPS-R@20 (%) Var$(\boldsymbol{w})$ KuaiRand LR-Restrict 0.74 2.80 7.50 12.47 0.83 KuaiRand LR-Full （我们的） 0.78 1.90 7.12 13.05 1.00 KuaiRand MLP-2layer 0.80 1.60 7.05 13.09 1.90 OBD LR-Restrict 0.76 2.50 6.50 12.00 0.78 OBD LR-Full（我们的） 0.80 1.70 6.20 12.74 1.00 OBD MLP-2layer 0.82 1.50 6.15 12.81 1.85

4.3. Temporal Consistency Contrast Without Heavy Augmentations 为了提高鲁棒性和长尾泛化能力，我们对会话表示施加了无视图一致性约束。对于每个会话 S，我们形成一个前缀 $S^{\text{pre}} = (x_1, \ldots, x_\tau)$ 和后缀 $S^{\text{suf}} = (x_{\tau+1}, \ldots, x_T)$ 与 $\tau \sim \text{Unif}\{2, \ldots, T-2\}$。将每个数据通过相同的编码器会产生嵌入 s^{pre}、$s^{\text{suf}} \in \mathbb{R}^d$。我们向两个视图添加一个小的潜在扰动$\epsilon \sim \mathcal{N}(0, \sigma^2 I)$以避免崩溃，同时避免结构图增强。直观上，这种损失鼓励编码器将同一会话的不同连续子序列（前缀和后缀）映射到表示空间中的附近点，https://doi.org/10.3390/ electronics15010084

while pushing apart representations of subsequences coming from different sessions. The Info Noise-Contrastive Estimation (InfoNCE) loss with in-batch negatives is

$$\mathcal{L}_{\text{con}} = \frac{1}{B}\sum_{i=1}^{B}\omega_i\left[-\log\frac{\exp\left(\langle\tilde{s}_i^{\text{pre}},\tilde{s}_i^{\text{suf}}\rangle/\tau\right)}{\sum_{j=1}^{B}\exp\left(\langle\tilde{s}_i^{\text{pre}},\tilde{s}_j^{\text{suf}}\rangle/\tau\right)}\right], \quad \tilde{s} = \frac{s+\epsilon}{\|s+\epsilon\|_2}. \tag{11}$$

Temporal structure is emphasized by a decay weight

$$\omega_i = \frac{\exp\left(-\eta\Delta t_i^{\text{gap}}\right)}{\frac{1}{B}\sum_{j=1}^{B}\exp\left(-\eta\Delta t_j^{\text{gap}}\right)}, \tag{12}$$

where Δt_i^{gap} is the time span between the prefix end and suffix start of session i. This weighting strengthens alignment when the two views are temporally close and relaxes it when they are far apart. Figure 3 shows the prefix–suffix construction and contrastive alignment.

The total training objective combines debiased supervision and temporal consistency:

$$\mathcal{J}(\theta) = \mathcal{J}_{\text{sup}}(\theta) + \lambda_{\text{con}}\mathcal{L}_{\text{con}}(\theta), \tag{13}$$

with $\lambda_{\text{con}} > 0$ tuned on a validation split.

Our prefix–suffix temporal consistency objective can be viewed as a form of consistency training on sequential data. Assume that each session S is generated from a latent intent variable z, and that both the prefix S^{pre} and suffix S^{suf} are conditionally independent given z. Under this assumption, an encoder f_θ that maps S to a representation $s = f_\theta(S)$ which is invariant across $(S^{\text{pre}}, S^{\text{suf}})$ pairs implicitly learns a sufficient statistic for z, in the sense that s preserves the information about the latent intent while discarding view-specific noise. Recent work on consistency training for sequential recommendation [33] and contrastive learning on event sequences [34] formalized similar intuitions, showing that enforcing agreement between multiple views (or subsequences) of the same sequence can reduce representation variance and improve generalization without requiring heavy augmentations. Our temporal consistency loss in Equation (11) follows this line of reasoning: it penalizes mutual information gaps between the representations of S^{pre} and S^{suf}, while our ablation results in Section 7 empirically confirm that adding TC on top of the SNIPS-based PA module consistently improves both unbiased accuracy and long-tail recall. We do not claim a new convergence theorem but rather place our design within this established theoretical framework and support it with targeted empirical evidence.

4.4. Training Algorithm and Complexity

The model is trained by mini-batch stochastic optimization over sessions. Within a batch, we compute session graphs, interval-aware attention, SNIPS weights, the debiased cross-entropy loss, and the contrastive loss. Inference uses the encoder once per session and a single linear projection to produce top-K items.

Algorithm 1 summarizes the training procedure.

Computational cost. For a session of length T, one graph-transformer layer attends over $|\mathcal{N}(i)| \leq 2$ neighbors when restricted to immediate transitions, yielding $O(HTd)$ complexity per layer, linear in session length. The propensity computation and SNIPS weighting are $O(B)$ per batch. The contrastive loss introduces an in-batch softmax of cost $O(Bd)$. The overall training throughput is dominated by encoder forward passes; memory scales linearly with T under per-session graphs.

同时推开来自不同会话的子序列的表示。批内负数的信息噪声对比估计 (InfoNCE) 损失为 $\mathcal{L}_{\text{con}} = \frac{1}{B}\sum_{i=1}^{B}\omega_i\left[-\log\frac{\exp(\langle \tilde{s}_i^{\text{pre}}, \tilde{s}_i^{\text{suf}}\rangle/\tau)}{\sum_{j=1}^{B}\exp(\langle \tilde{s}_i^{\text{pre}}, \tilde{s}_j^{\text{suf}}\rangle/\tau)}\right]$，$\tilde{s} = \frac{s+\epsilon}{\|s+\epsilon\|_2}$。 (11) 时间结构通过衰减权重 $\omega_i = \frac{\exp(-\eta\Delta t_i^{\text{gap}})}{\sum_{j=1}^{B}\exp(-\eta\Delta t_j^{\text{gap}})}$ 来强调， (12) 其中 Δt_i^{gap} 是会话 i 的前缀结束和后缀开始之间的时间跨度。当两个视图在时间上接近时，这种加权会加强对齐，而当它们相距较远时，会放松对齐。图 3 显示了前缀-后缀结构和对比对齐。总的训练目标结合了去偏监督和时间一致性：

$\mathcal{J}(\theta) = \mathcal{J}_{\text{sup}}(\theta) + \lambda_{\text{con}}\mathcal{L}_{\text{con}}(\theta)$, (13) 和 $\lambda_{\text{con}} > 0$ 在验证分割上调整。我们的前缀-后缀时间一致性目标可以被视为对顺序数据进行一致性训练的一种形式。假设每个会话 S 都是从潜在意图变量 z 生成的，并且前缀 S^{pre} 和后缀 S^{suf} 在给定 z 的情况下是条件独立的。在此假设下，将 S 映射到在 $(S^{\text{pre}}, S^{\text{suf}})$ 对之间不变的表示 $s = f_\theta(S)$ 的编码器 f_θ 隐式地学习了 z 的足够统计量，从某种意义上说，s 保留了有关潜在意图的信息，同时丢弃了特定于视图的噪声。最近关于顺序推荐的一致性训练[33]和事件序列的对比学习[34]形式化了类似的直觉，表明在同一序列的多个视图（或子序列）之间强制达成一致可以减少表示方差并提高泛化性，而无需大量增强。我们在等式（11）中的时间一致性损失遵循这样的推理：它惩罚了 S^{pre} 和 S^{suf} 表示之间的相互信息差距，而我们在第 7 节中的消融结果凭经验证实，在基于 SNIPS 的 PA 模块上添加 TC 一致提高了无偏准确度和长尾召回率。我们并不主张新的收敛定理，而是将我们的设计置于这个既定的理论框架内，并用有针对性的经验证据来支持它。 *4.4. Training Algorithm and Complexity* 该模型通过会话的小批量随机优化进行训练。在一个批次中，我们计算会话图、间隔感知注意力、SNIPS 权重、去偏交叉熵损失和对比损失。推理每个会话使用一次编码器和单个线性投影来生成 top-K 项。算法 1 总结了训练过程。计算成本。对于长度为 T 的会话，当限制为立即转换时，一个图变换器层会参与超过 $|\mathcal{N}(i)| \leq 2$ 个邻居，从而每层产生 $O(HTd)$ 复杂性，与会话长度呈线性关系。每批次的倾向计算和 SNIPS 权重为 $O(B)$。对比损失引入了成本 $O(Bd)$ 的批量 softmax。整体训练吞吐量由编码器前向传递主导；在每个会话图表下，内存与 T 线性缩放。

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Algorithm 1 Debiased session-graph training with temporal consistency.

Inputs: training logs D, randomized slice R, rank-wise random rates $\{\rho_k\}$, encoder $f_\theta(\cdot)$, exposure model $q_\phi(\cdot)$, temperature τ, contrast weight λ_{con}, clip c, variance λ.
Fit exposure model q_ϕ on R by maximizing likelihood of observed displays.
for each training epoch **do**
  for each mini-batch $B = \{(S, y, a, k)\}$ **do**
    ▷ Forward pass
    $s \leftarrow f_\theta(S)$ ▷ session embedding via interval-aware graph transformer
    $p \leftarrow \text{softmax}(W_o s)$ ▷ next-item distribution
    ▷ Propensity weights
    $\hat{p} \leftarrow z \cdot \rho_k + (1 - z) \cdot q_\phi(S, y, k)$ ▷ where z indicates randomized assignment
    $w \leftarrow \min(c, 1/\hat{p})$; $\hat{w} \leftarrow w/\text{mean}(w)$
    $\mathcal{L}_{\text{sup}} \leftarrow (\sum \hat{w} \cdot \text{CE}(p, y))/(\sum \hat{w}) + \lambda \cdot \text{Var}(w)$
    ▷ Temporal consistency
    sample τ uniformly; form prefix S_{pre} and suffix S_{suf} from S
    $s_{\text{pre}} \leftarrow f_\theta(S_{\text{pre}})$; $s_{\text{suf}} \leftarrow f_\theta(S_{\text{suf}})$
    add Gaussian noise to $s_{\text{pre}}, s_{\text{suf}}$; $\ell_{\text{con}} \leftarrow \text{InfoNCE}(s_{\text{pre}}, s_{\text{suf}}; \tau)$
    ▷ Total loss and update
    $\mathcal{L} \leftarrow \mathcal{L}_{\text{sup}} + \lambda_{\text{con}} \cdot \ell_{\text{con}}$
    update θ by AdamW on $\nabla_\theta \mathcal{L}$
  end for
end for

5. Experimental Setup

5.1. Dataset Preprocessing and Evaluation Metrics

We follow the standard public preprocessing protocols for each dataset. For YOOCHOOSE-1/64, this involves filtering out sessions of length one and applying prefix expansion to create labeled sequences.

Our primary evaluation metrics are Recall@K, Normalized Discounted Cumulative Gain (NDCG@K), and Mean Reciprocal Rank (MRR@K), where we report K = 10 and K = 20. In our results tables, we abbreviate Recall@K, NDCG@K, and MRR@K as R@K, N@K, and M@K, respectively, for brevity. For the OTTO dataset, we also report the official single-objective click metrics to ensure a fair comparison with published baselines. On datasets containing randomized interventions or logged propensities (KuaiRand and OBD), we conduct an unbiased evaluation, reporting metrics weighted by SNIPS. We denote the SNIPS-weighted Recall@K and MRR@K as SNIPS-R@K and SNIPS-M@K. On standard observational datasets (OTTO and YOOCHOOSE), we report the standard (biased) top-K metrics. Beyond accuracy, we analyze model calibration using the Expected Calibration Error (ECE) and Brier Score, and assess performance on long-tail item distributions.

For OBD, we use the "all" campaign, filter impressions where the click indicator is defined, and treat each impression as a training instance. We use the provided `propensity_score` field as the behavior-policy exposure probability for both training (SNIPS-weighted objective) and evaluation (SNIPS-weighted metrics). As OBD is impression-based rather than session-based, we adapt next-item prediction to the impression scenario by conditioning on short impression contexts when applicable and aligning the evaluation protocol with off-policy evaluation practice.

Unless otherwise stated, all reported results are averaged over five random seeds. For clarity of presentation, we report mean values in the main performance and unbiased evaluation tables and summarize standard deviations and statistical significance for key comparisons in a dedicated table devoted to statistical tests. We compute paired t-tests for standard metrics and cluster bootstrap confidence intervals for SNIPS-weighted metrics between PTE-GT and the strongest baseline on each dataset; the improvements of PTE-GT

 算法 1 具有时间一致性的去偏会话图训练。 1：输入：训练日志 D、随机切片 R、按等级随机率 $\{\rho_k\}$、编码器 $f_\theta(\cdot)$、曝光模型 $q_\phi(\cdot)$、温度 τ、对比度权重 λ_{con}、剪辑 c、方差 λ。 2：通过最大化观察到的显示的可能性，在 R 上拟合曝光模型 q_ϕ。 3：对于每个训练时期执行 4：对于每个小批量 $B = \{(S、y、a、k)\}$ 执行 5：▷ 前向传递 6：通过间隔感知图形转换器进行 $s \leftarrow f_\theta(S)$ ▷ 会话嵌入 7：$p \leftarrow \text{softmax}(W_o s)$ ▷ 下一项分布 8: ▷ 倾向权重 9: $\hat{p} \leftarrow z \cdot \rho_k + (1 - z) \cdot q_\phi(S、y、k)$ ▷，其中 z 表示随机分配 10: $w \leftarrow \min(c, 1/\hat{p})$；$\hat{w} \leftarrow w/\text{mean}(w)$ 11: $\mathcal{L}_{\text{sup}} \leftarrow (\sum \hat{w} \cdot \text{CE}(p, y))/(\sum \hat{w}) + \lambda \cdot \text{Var}(w)$ 12: ▷ 时间一致性 13: 均匀采样 τ；从 S 14: $s_{\text{pre}} \leftarrow f_\theta(S_{\text{pre}})$ 形成前缀 S_{pre} 和后缀 S_{suf}； $s_{\text{suf}} \leftarrow f_\theta(S_{\text{suf}})$ 15: 将高斯噪声添加到 s_{pre}、s_{suf} 中； $\ell_{\text{con}} \leftarrow \text{InfoNCE}(s_{\text{pre}}, s_{\text{suf}}; \tau)$ 16: ▷ Total loss and update 17: $\mathcal{L} \leftarrow \mathcal{L}_{\text{sup}} + \lambda_{\text{con}} \cdot \ell_{\text{con}}$ update θ by AdamW on $\nabla_\theta \mathcal{L}$ 19: end for 20: end for 18:

5. 实验设置

5.1. Dataset Preprocessing and Evaluation Metrics

我们遵循每个数据集的标准公共预处理协议。对于 YOOCHOOSE-1/64，这涉及过滤掉长度为 1 的会话并应用前缀扩展来创建标记序列。我们的主要评估指标是 Recall@K、归一化贴现累积增益 (NDCG@K) 和平均倒数排名 (MRR@K)，其中我们报告 K = 10 和 K = 20。在我们的结果表中，为简洁起见，我们将 Recall@K、NDCG@K 和 MRR@K 分别缩写为 R@K、N@K 和 M@K。对于 OTTO 数据集，我们还报告了官方单目标点击指标，以确保与已发布的基线进行公平比较。在包含随机干预或记录倾向（KuaiRand 和 OBD）的数据集上，我们进行公正的评估，报告由 SNIPS 加权的指标。我们将 SNIPS 加权 Recall@K 和 MRR@K 表示为 SNIPS-R@K 和 SNIPS-M@K。在标准观测数据集（OTTO 和 YOOCHOOSE）上，我们报告标准（有偏差的）top-K 指标。除子准确性之外，我们还使用预期校准误差 (ECE) 和 Brier 分数来分析模型校准，并评估长尾项目分布的性能。对于OBD，我们使用“全部”活动，过滤定义了点击指示器的展示次数，并将每个展示次数视为一个训练实例。我们使用提供的 propensity_score 字段作为训练（SNIPS 加权目标）和评估（SNIPS 加权指标）的行为策略暴露概率。由于 OBD 是基于印象而不是基于会话的，因此我们通过在适用时调节短印象上下文并使评估协议与非策略评估实践保持一致，使下一项预测适应印象场景。除非另有说明，所有报告的结果都是五个随机种子的平均值。为了清楚地表达，我们报告了主要性能和无偏评估表中的平均值，并在专门用于统计测试的专用表中总结了关键比较的标准差和统计显着性。我们计算标准指标的配对 t 测试，以及 PTE-GT 和每个数据集上最强基线之间 SNIPS 加权指标的集群引导置信区间； PTE-GT https://doi.org/10.3390/electronics15010084 的改进

over TRON on OTTO and over IGT-Base on KuaiRand are highly significant ($p < 10^{-4}$), while the gains on YOOCHOOSE and OBD are also statistically significant with $p < 0.05$.

5.2. Baselines and Implementation Details

We compare our proposed PTE-GT model against a suite of strong baselines, including: (1) GRU4Rec+, a widely cited RNN-based model [41]; (2) SASRec, a representative Transformer-based sequential model [5]; (3) SR-GNN, a canonical GNN-based session model [23]; (4) TRON, a strong industrial Transformer baseline from the OTTO challenge [24]; and (5) IGT-Base, our backbone interval-aware graph transformer trained with standard cross-entropy loss [25]. For the unbiased debiasing comparison on KuaiRand and OBD, we additionally consider two causal/DR-style baselines: (6) CausalRec-adapt, which follows the front-door debiasing idea of CausalRec [35] but uses SASRec as a backbone and replaces visual features with item popularity and category features as proxies for exposure; and (7) DR-JL, a doubly robust learning variant that combines IPS with a separate reward model in the spirit of DR estimators [36,37], again instantiated on top of SASRec.

Software and Computing Environment

All experiments were implemented in Python (3.11) and PyTorch (2.9.0). Training was conducted on an NVIDIA A100 GPU with CUDA (11.1) and cuDNN (v8.9.7).

All models are implemented in PyTorch and trained on a single NVIDIA A100 graphics processing unit (GPU). We use the AdamW optimizer for all models. The learning rate is selected via grid search from $\{1 \times 10^{-3}, 5 \times 10^{-4}, 1 \times 10^{-4}\}$ and the batch size is set to 512. The item and position embedding dimension is set to 128 across all models for fair comparison. For our PTE-GT model, the IGT backbone uses $L = 2$ graph transformer layers and $H = 4$ attention heads. Based on validation set tuning, the SNIPS clipping threshold c is set to 20, and the temporal consistency temperature τ is set to 0.1.

6. Results

This section details our experimental results on four datasets, including three public session-based benchmarks—OTTO, YOOCHOOSE-1/64, and KuaiRand—and the Open Bandit Dataset (OBD). We compare our proposed model, PTE-GT, with a series of strong baselines.

6.1. Main Performance on Public Benchmarks

We first evaluate the standard top-K recommendation accuracy of our models on two widely used public benchmarks, OTTO and YOOCHOOSE-1/64. OTTO represents a large-scale, sparse, real-world e-commerce scenario, while YOOCHOOSE-1/64 is a smaller, standard benchmark widely used for comparison. Results are summarized in Table 3.

The results in Table 3 demonstrate the effectiveness of PTE-GT on both datasets. IGT-Base refers to using only our interval-aware graph transformer backbone without propensity-aware training or temporal consistency regularization. On the YOOCHOOSE-1/64 dataset, all graph- and Transformer-based models, including SR-GNN, TRON, and IGT-Base, perform strongly, significantly outperforming GRU4Rec. Our IGT-Base backbone alone achieves state-of-the-art performance comparable to TRON. After adding our propensity-aware and temporal-consistency modules, our full model, PTE-GT, achieves the best performance on both R@20 and M@20, validating our approach on this standard benchmark.

 超过 OTTO 上的 TRON 和 KuaiRand 上超过 IGT-Base 的收益非常显着（$p < 10^{-4}$），而 YOOCHOOSE 和 OBD 的收益也具有统计显着性，$p < 0.05$。

*5.2. Baselines and Implementation Details*我们将我们提出的 PTE-GT 模型与一组强大的基线进行比较，包括：（1）GRU4Rec+，一种被广泛引用的基于 RNN 的模型 [41]； (2)SASRec，一种代表性的基于Transformer的序列模型[5]； (3) SR-GNN，一种基于 GNN 的规范会话模型 [23]； (4) TRON，来自 OTTO 挑战的强大工业 Transformer 基线 [24]； (5) IGT-Base，我们的骨干区间感知图转换器，经过标准交叉熵损失训练[25]。对于 KuaiRand 和 OBD 上的无偏去偏比较，我们还考虑了两个因果/DR 风格的基线：（6）CausalRec-adapt，它遵循 CausalRec [35] 的前门去偏思想，但使用 SASRec 作为骨干，并用项目流行度和类别特征替换视觉特征作为曝光代理； (7) DR-JL，一种双重稳健的学习变体，本着 DR 估计器的精神将 IPS 与单独的奖励模型结合起来 [36,37]，再次在 SASRec 之上实例化。软件和计算环境所有实验均在Python（3.11）和PyTorch（2.9.0）中实现。训练是在配备 CUDA (11.1) 和 cuDNN (v8.9.7) 的 NVIDIA A100 GPU 上进行的。所有模型均在 PyTorch 中实现，并在单个 NVIDIA A100 图形处理单元 (GPU) 上进行训练。我们对所有模型使用 AdamW 优化器。学习率通过网格搜索从 $\{1 \times 10^{-3}$、5×10^{-4}、$1 \times 10^{-4}\}$ 中选择，批量大小设置为 512。为了公平比较，所有模型的项目和位置嵌入维度设置为 128。对于我们的 PTE-GT 模型，IGT 主干使用 $L = 2$ 个图形转换器层和 $H = 4$ 个注意力头。基于验证集调整，SNIPS 裁剪阈值 c 设置为 20，时间一致性温度 τ 设置为 0.1。 6.结果本节详细介绍了我们在四个数据集上的实验结果，包括三个基于公共会话的基准——OTTO、YOOCHOOSE-1/64和KuaiRand——以及开放强盗数据集（OBD）。我们将我们提出的模型 PTE-GT 与一系列强大的基线进行比较。 *6.1. Main Performance on Public Benchmarks* 我们首先在两个广泛使用的公共基准 OTTO 和 YOOCHOOSE-1/64 上评估我们模型的标准 top-K 推荐准确性。 OTTO 代表了大规模、稀疏的真实电子商务场景，而 YOOCHOOSE-1/64 是一个较小的、广泛用于比较的标准基准。结果总结于表 3 中。表 3 中的结果证明了 PTE-GT 在两个数据集上的有效性。 IGT-Base 是指仅使用我们的区间感知图转换器主干，而不使用倾向感知训练或时间一致性正则化。在 YOOCHOOSE-1/64 数据集上，所有基于图和 Transformer 的模型，包括 SR-GNN、TRON 和 IGT-Base，都表现强劲，显着优于 GRU4Rec。仅我们的 IGT-Base 主干网就实现了与 TRON 相当的最先进的性能。添加倾向感知和时间一致性模块后，我们的完整模型 PTE-GT 在 R@20 和 M@20 上实现了最佳性能，验证了我们在此标准基准上的方法。

https://doi.org/10.3390/electronics15010084

Table 3. Comprehensive performance and complexity comparison on OTTO and YOOCHOOSE-1/64 datasets (%). All results are averaged over five random seeds.

Model	Params (M)	OTTO (Clicks)						YOOCHOOSE-1/64					
		R@10	N@10	M@10	R@20	N@20	M@20	R@10	N@10	M@10	R@20	N@20	M@20
Baselines (Reproduced/Literature)													
GRU4Rec+	5.2	27.12	14.03	13.51	44.31	27.84	20.53	38.17	25.02	15.11	60.63	34.16	22.89
SASRec	6.8	19.04	11.52	11.03	30.71	20.13	18.02	45.91	32.14	19.06	69.17	40.82	29.88
SR-GNN	7.1	28.03	14.81	14.12	45.83	28.91	21.14	48.02	34.89	20.18	70.56	42.11	30.94
TRON	7.0	29.01	15.53	14.92	47.24	30.03	21.91	48.53	35.11	20.52	70.83	42.51	31.02
Ours													
IGT-Base	7.5	29.54	15.92	15.21	47.91	30.52	22.34	48.71	35.26	20.61	70.97	42.63	31.07
PTE-GT (Full)	7.7	30.53	16.64	15.82	49.15	31.54	23.01	49.84	36.23	21.42	71.86	43.82	32.05
Improv. (vs. TRON)	-	+5.24%	+7.15%	+6.03%	+4.04%	+5.03%	+5.02%	+2.70%	+3.19%	+4.39%	+1.45%	+3.08%	+3.32%

On the much larger and more challenging OTTO dataset, this advantage is more pronounced. TRON is the strong baseline reported officially. Our IGT-Base backbone again slightly outperforms TRON. Our full model, PTE-GT, achieves a significant performance improvement, with a 4.0% relative increase in R@20 and a 5.0% relative increase in M@20 over TRON. This indicates that our training strategy is robust in handling the sparsity and inherent biases of large-scale, real-world data.

6.2. Unbiased Evaluation on Randomized/Logged-Propensity Datasets

Standard evaluation (as in Table 3) is conducted on observational logs, which can be influenced by exposure bias, thereby inflating performance. To more accurately measure the true performance of the models in an unbiased setting, we conduct unbiased evaluation on KuaiRand-27K and the Open Bandit Dataset (OBD). KuaiRand contains randomly exposed interactions, while OBD provides logged behavior-policy propensities, enabling us to use SNIPS to estimate unbiased evaluation metrics on both datasets. Table 4 summarizes the SNIPS-weighted Recall@20 and MRR@20 on these two datasets.

Table 4. Unbiased evaluation on KuaiRand and OBD (SNIPS-weighted, %). All metrics are averaged over five random seeds.

Model	KuaiRand-27K		OBD (All)	
	SNIPS-R@20 (%)	SNIPS-M@20 (%)	SNIPS-R@20 (%)	SNIPS-M@20 (%)
SASRec	10.15	4.32	9.92	4.18
IGT-Base	10.88	4.65	10.63	4.44
SASRec-IPS	11.75	5.02	11.48	4.87
CausalRec-adapt	12.03	5.21	11.96	5.04
DR-JL	12.54	5.32	12.31	5.17
PTE-GT (Full Model)	13.05	5.58	12.74	5.36

The first three rows in Table 4 correspond to "naive" models trained with standard (unweighted) objectives: SASRec and IGT-Base both suffer from severe exposure bias, and their SNIPS-R@20 scores are noticeably lower than those of their debiased counterparts. Adding IPS weighting on top of SASRec (SASRec-IPS) already improves the unbiased metrics, confirming the effectiveness of classical off-policy correction.

We then include two stronger debiasing baselines that are closer to recent causal recommendation work. CausalRec-adapt follows the front-door style debiasing idea of CausalRec but replaces visual features with item popularity and category features as proxies for exposure, and uses SASRec as the rating backbone. DR-JL is a doubly robust variant that combines IPS with a learned reward model, instantiated on top of SASRec, inspired

Electronics 2026, *15*, 84 13 of 26 表 3. OTTO 和 YOOCHOOSE-1/64 数据集的综合性能和复杂性比较 (%)。所有结果都是五个随机种子的平均值。模型参数 (M) OTTO（点击次数） YOOCHOOSE-1/64 R@10 N@10 M@10 R@20 N@20 M@20 R@10 N@10 M@10 R@20 N@20 M@20 *Baselines (Reproduced/Literature)* GRU4Rec+ 5.2 27.12 14.03 13.51 44.31 27.84 20.53 38.17 25.02 15.11 60.63 34.16 22.89 SASRec 6.8 19.04 11.52 11.03 30.71 20.13 18.02 45.91 32.14 19.06 69.17 40.82 29.88 SR-GNN 7.1 28.03 14.81 14.12 45.83 28.91 21.14 48.02 34.89 20.18 70.56 42.11 30.94 波场 7.0 29.01 15.53 14.92 47.24 30.03 21.91 48.53 35.11 20.52 70.83 42.51 31.02 *Ours* IGT 基 7.5 29.54 15.92 15.21 47.91 30.52 22.34 48.71 35.26 20.61 70.97 42.63 31.07 PTE-GT（完全） 7.7 30.53 16.64 15.82 49.15 31.54 23.01 49.84 36.23 21.42 71.86 43.82 32.05 改进（对比波场）- +5.24% +7.15% +6.03% +4.04% +5.03% +5.02% +2.70% +3.19% +4.39% +1.45% +3.08% +3.32% 在更大、更具挑战性的 OTTO 数据集，这种优势更加明显。 TRON 是官方报告的强基线。我们的 IGT-Base 主干网的性能再次略胜于 TRON。我们的完整模型 PTE-GT 实现了显着的性能提升，与 TRON 相比，R@20 相对增加了 4.0%，M@20 相对增加了 5.0%。这表明我们的训练策略在处理大规模真实世界数据的稀疏性和固有偏差方面是稳健的。

6.2. Unbiased Evaluation on Randomized/Logged-Propensity Datasets 标准评估（如表 3 所示）是根据观察日志进行的，这可能会受到曝光偏差的影响，从而夸大性能。为了更准确地衡量模型在无偏差环境下的真实性能，我们对 KuaiRand-27K 和开放 Bandit 数据集（OBD）进行无偏差评估。 KuaiRand 包含随机暴露的交互，而 OBD 提供记录的行为策略倾向，使我们能够使用 SNIPS 来估计两个数据集上的无偏评估指标。表 4 总结了这两个数据集上的 SNIPS 加权 Recall@20 和 MRR@20。表 4. 对 KuaiRand 和 OBD 的公正评价（SNIPS 加权，%）。所有指标都是五个随机种子的平均值。型号 KuaiRand-27K OBD (全部) SNIPS-R@20 (%) SNIPS-M@20 (%) SNIPS-R@20 (%) SNIPS-M@20 (%) SASRec 10.15 4.32 9.92 4.18 IGT-Base 10.88 4.65 10.63 4.44 SASRec-IPS 11.75 5.02 11.48 4.87 CausalRec-adapt 12.03 5.21 11.96 5.04 DR-JL 12.54 5.32 12.31 5.17 PTE-GT (完整模型) 13.05 5.58 12.74 5.36 表 4 中的前三行对应到使用标准（未加权）目标训练的“朴素”模型：SASRec 和 IGT-Base 都遭受严重的暴露偏差，并且它们的 SNIPS-R@20 分数明显低于除偏差同行的分数。在 SASRec (SASRec-IPS) 之上添加 IPS 权重已经改善了无偏指标，证实了经典离策略校正的有效性。然后，我们包括两个更接近最近因果推荐工作的更强的去偏基线。 CausalRec-adapt 遵循 CausalRec 的前门风格去偏思想，但用项目流行度和类别特征替换视觉特征作为曝光代理，并使用 SASRec 作为评级主干。 DR-JL 是一种双重稳健的变体，它将 IPS 与学习奖励模型相结合，在 SASRec 之上实例化，灵感来自 https://doi.org/10.3390/ electronics15010084

by DR and multiple robust estimators [36,37]. Both methods outperform SASRec-IPS on KuaiRand and OBD; for instance, DR-JL reaches 12.54% SNIPS-R@20 on KuaiRand and 12.31% on OBD.

However, our full PTE-GT model still achieves the highest SNIPS-weighted accuracy on both datasets: 13.05% SNIPS-R@20 on KuaiRand and 12.74% on OBD, together with the best SNIPS-M@20. This shows that the gains of PTE-GT are not merely due to "adding IPS" but stem from the combination of an interval-aware graph transformer backbone, a carefully calibrated SNIPS objective, and the temporal consistency regularizer. The gaps between PTE-GT and the causal/DR baselines are substantially larger under unbiased evaluation than those observed on standard OTTO metrics, highlighting again that conventional offline evaluation tends to underestimate the value of principled debiasing.

6.3. Analysis of Recommendation Calibration

Beyond accuracy, the reliability of the predicted probabilities output by a recommender system (i.e., calibration) is also critical. A well-calibrated model's predicted confidence (probability) should align with its actual accuracy. We use ECE and Brier Score to quantify model calibration, where lower is better for both. As shown in Table 5, the IGT-Base model trained without debiasing exhibits poor calibration (higher ECE) on both datasets. This is expected, as the model tends to be overconfident in its predictions for over-exposed popular items. In contrast, our PTE-GT model, through propensity-aware training, significantly reduces both ECE and Brier Score. For example, on the OTTO dataset, ECE was reduced from 5.12% to 2.85%, a 44.3% reduction.

Table 5. Calibration performance on OTTO and KuaiRand (ECE (%), Brier (%)).

Model	OTTO (Standard)		KuaiRand (Unbiased)	
	ECE (%)	Brier (%)	ECE (%)	Brier (%)
IGT-Base	5.12	7.82	6.45	9.18
PTE-GT	2.85	5.11	3.11	6.24

The reliability diagrams in Figure 4a,c plot the empirical accuracy (y-axis) against the model's average predicted probability (x-axis) for binned predictions. For IGT-Base, the curve consistently falls well below the perfect calibration diagonal, confirming the significant overconfidence first identified in Table 5. In stark contrast, the PTE-GT curve tracks the diagonal much more closely on both KuaiRand and OTTO, demonstrating superior calibration.

The confidence histograms in Figure 4b,d reveal the cause of this miscalibration. IGT-Base exhibits a common symptom of overconfidence: it pushes a large, unhealthy proportion of its predictions into the highest confidence bin (e.g., 0.9–1.0). However, as shown in the reliability plots in Figure 4a,c, its actual accuracy in this bin is far lower (e.g., 70–78%). Our PTE-GT model corrects this, pulling these overconfident predictions back and redistributing them into more appropriate, lower-confidence bins. This improved calibration is crucial for building trust and for downstream tasks like balancing exploration and exploitation in a production environment.

6.4. Performance on Long-Tail Items

Finally, we conduct a comprehensive analysis of the models' performance on items of varying popularity, from popular "head" items to "long-tail" items. Exposure bias typically causes models to overfit to popular items, neglecting the long tail. We investigate this by evaluating performance across 10 item popularity deciles (decile 1 = top 10% most popular, decile 10 = 10% least popular).

 by DR 和多个稳健估计器 [36,37]。两种方法在 KuaiRand 和 OBD 上均优于 SASRec-IPS；例如，DR-JL在快兰德上达到SNIPS-R@20的12.54%，在OBD上达到12.31%。然而，我们的完整 PTE-GT 模型仍然在两个数据集上实现了最高的 SNIPS 加权精度：KuaiRand 上的 SNIPS-R@20 为 13.05%，OBD 上为 12.74%，以及最好的 SNIPS-M@20。这表明 PTE-GT 的增益不仅仅是由于“添加 IPS”，而是源于区间感知图转换器主干、仔细校准的 SNIPS 目标和时间一致性正则化器的组合。在无偏评估下，PTE-GT 与因果/DR 基线之间的差距比在标准 OTTO 指标上观察到的差距要大得多，这再次凸显出传统的离线评估往往低估了原则性去偏的价值。

6.3. Analysis of Recommendation Calibration 除了准确性之外，推荐系统输出的预测概率（即校准）的可靠性也至关重要。校准良好的模型的预测置信度（概率）应与其实际准确性保持一致。我们使用 ECE 和 Brier Score 来量化模型校准，两者都越低越好。如表 5 所示，未经去偏训练的 IGT-Base 模型在两个数据集上都表现出较差的校准（较高的 ECE）。这是预料之中的，因为该模型往往对过度曝光的热门商品的预测过于自信。相比之下，我们的 PTE-GT 模型通过倾向感知训练，显着降低了 ECE 和 Brier 分数。例如，在 OTTO 数据集上，ECE 从 5.12% 降低到 2.85%，降低了 44.3%。表 5. OTTO 和 KuaiRand 上的校准性能（ECE (%)、Brier (%)）。模型 OTTO（标准） KuaiRand（无偏） ECE (%) Brier (%) ECE (%) Brier (%) IGT-Base 5.12 7.82 6.45 9.18 PTE-GT 2.85 5.11 3.11 6.24 图 4a、c 中的可靠性图绘制了经验精度（y 轴）与模型平均预测概率的关系（x 轴）用于分箱预测。对于 IGT-Base，曲线始终远低于完美校准对角线，证实了表 5 中首次发现的严重过度自信。形成鲜明对比的是，PTE-GT 曲线在 KuaiRand 和 OTTO 上更接近地跟踪对角线，证明了卓越的校准。图 4b、d 中的置信直方图揭示了这种校准错误的原因。 IGT-Base 表现出过度自信的常见症状：它将大部分不健康的预测推入最高置信区间（例如 0.9-1.0）。然而，如图 4a、c 中的可靠性图所示，该箱中的实际准确度要低得多（例如 70-78%）。我们的 PTE-GT 模型纠正了这一问题，将这些过度自信的预测拉回来，并将它们重新分配到更合适、置信度较低的分类中。这种改进的校准对于建立信任和平衡生产环境中的探索和利用等下游任务至关重要。 *6.4. Performance on Long-Tail Items* 最后，我们对模型在不同受欢迎程度的商品（从流行的“头部”商品到“长尾”商品）上的表现进行了全面分析。曝光偏差通常会导致模型过度拟合流行商品，而忽略了长尾。我们通过评估 10 个项目流行度十分位（十分位 1 = 前 10% 最受欢迎，十分位 10 = 10% 最不受欢迎）的表现来调查这一点。 https://doi.org/10.3390/ electronics15 010084

The results are presented in the 2 × 4 matrix in Figure 5, which examines both Recall@20 for discovery breadth and MRR@20 for ranking quality across all four datasets.

The analysis reveals two critical findings. First, on all datasets, the baseline models (SASRec, IGT-Base) exhibit a sharp performance drop-off as items become less popular, confirming they are dominated by popularity bias. Our PTE-GT model consistently maintains a flatter curve, demonstrating superior performance on long-tail items (deciles 6–10).

Second, and most importantly, the figure highlights the limitations of standard (biased) evaluations. On OTTO and YOOCHOOSE (Figure 5a,b,e,f), the performance gap, while consistent, appears modest. However, under the principled, unbiased SNIPS evaluation on KuaiRand and OBD (Figure 5c,d,g,h), this gap widens substantially, especially in the long-tail deciles. Taken together, these results suggest that relying solely on observational logs can underestimate the benefits of debiasing, particularly for infrequent items.

This comprehensive 8-plot analysis indicates that our propensity-aware objective and temporal consistency regularization work in synergy, effectively mitigating popularity bias to enhance both recall-driven discovery and MRR-based ranking of long-tail items.

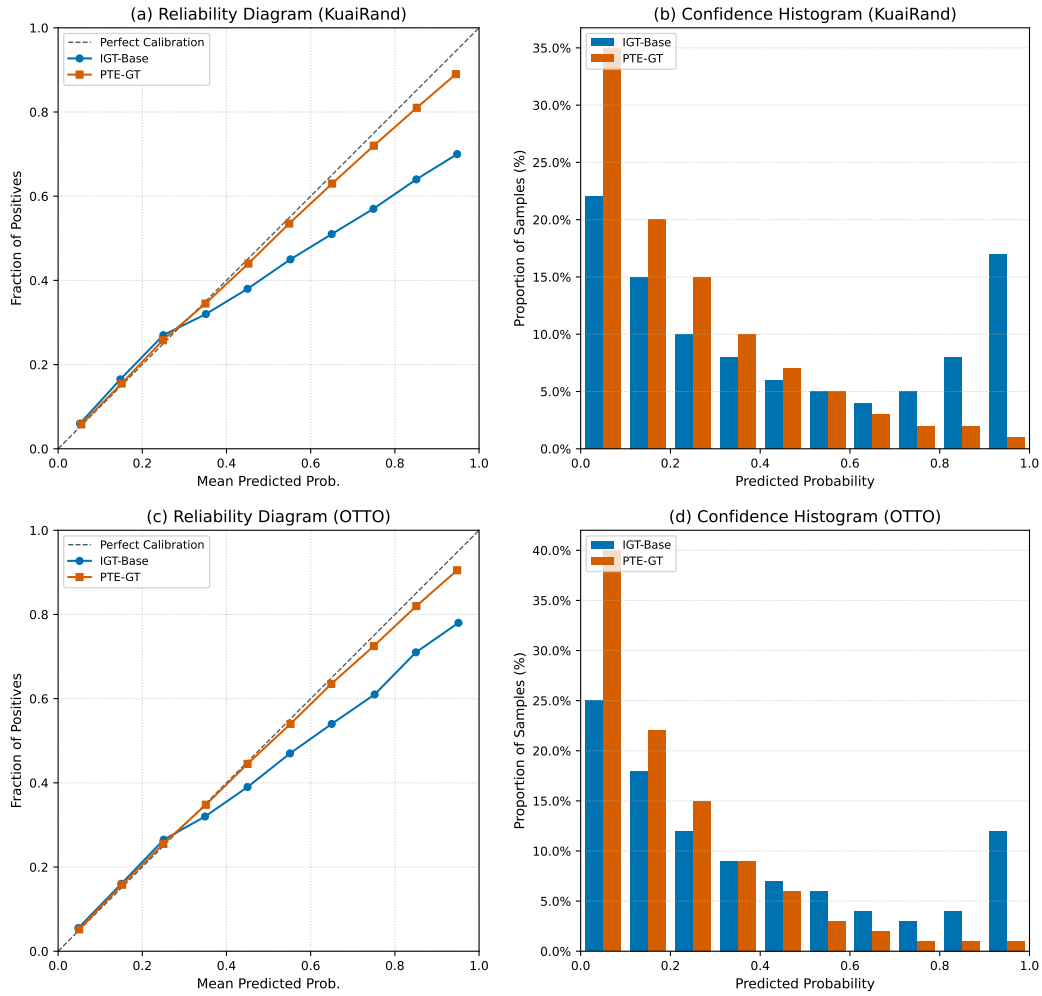
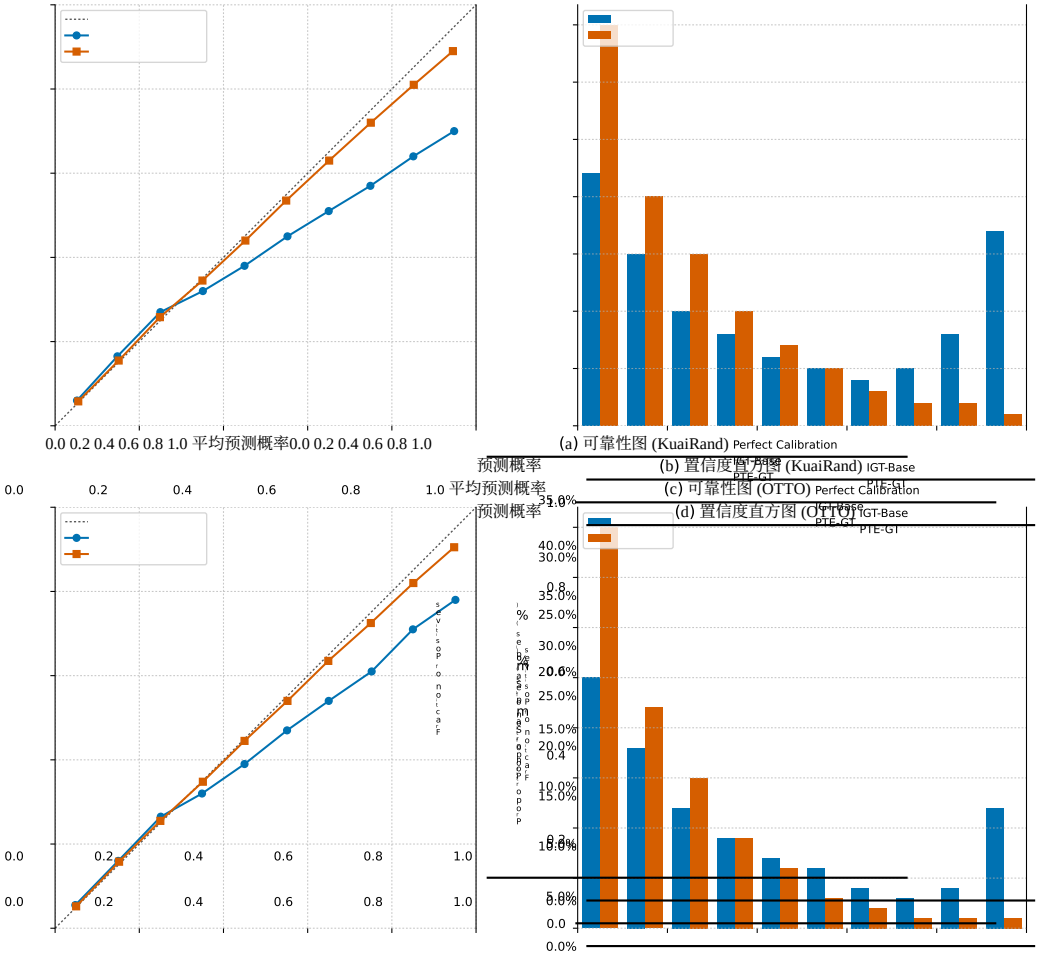


Figure 4. Comprehensive calibration analysis on KuaiRand and OTTO. (**a**,**c**) Reliability diagrams comparing IGT-Base and PTE-GT; the diagonal (dashed line) represents perfect calibration. (**b**,**d**) Histograms of predicted confidence for ground-truth items, illustrating that PTE-GT avoids the overconfidence of the baseline.

To provide a finer-grained view of debiased performance on datasets with unbiased propensities, Figure 6 plots SNIPS-weighted Recall@20 and MRR@20 by item popularity decile for KuaiRand and OBD, comparing IGT-Base and PTE-GT. On KuaiRand, PTE-GT

 结果显示在图 5 中的 2 × 4 矩阵中，该矩阵检查了 Recall@20 的发现广度和 MRR@20 的所有四个数据集的质量排名。分析揭示了两个重要发现。首先，在所有数据集上，随着项目变得不那么受欢迎，基线模型（SASRec、IGT-Base）表现出急剧的性能下降，证实它们受到流行度偏差的主导。我们的 PTE-GT 模型始终保持较平坦的曲线，在长尾项目（十分位数 6-10）上表现出卓越的性能。其次，也是最重要的一点，该图凸显了标准（有偏见的）评估的局限性。在 OTTO 和 YOOCHOOSE 上（图 5a、b、e、f），性能差距虽然一致，但似乎不大。然而，根据对 KuaiRand 和 OBD 原则性、公正的 SNIPS 评估（图 5c、d、g、h），这一差距大幅扩大，尤其是在长尾十分位数方面。总而言之，这些结果表明，仅依赖观察日志可能会低估去偏的好处，特别是对于不常见的项目。这种全面的 8 图分析表明，我们的倾向感知客观性和时间一致性正则化可以协同工作，有效地减轻流行度偏差，从而增强召回驱动的发现和基于 MRR 的长尾项目排名。图 4. KuaiRand 和 OTTO 的综合校准分析。 (a,c) 比较 IGT-Base 和 PTE-GT 的可靠性图；对角线（虚线）代表完美校准。 (b,d) 真实项目的预测置信度直方图，说明 PTE-GT 避免了基线的过度自信。为了提供具有无偏倾向的数据集上的去偏性能的更细粒度视图，图 6 按项目流行度十分位数绘制了 KuaiRand 和 OBD 的 SNIPS 加权 Recall@20 和 MRR@20，比较了 IGT-Base 和 PTE-GT。在快兰德上，PTE-GT https://doi.org/10.3390/ electronics15010084



improves SNIPS-R@20 from 18.3% to 18.9% in the head decile (1), but the relative gains grow steadily toward the tail, reaching +1.4 percentage points in the bottom-10% items (3.6% to 5.0%, approximately +39%). The corresponding SNIPS-M@20 in the tail rises from 1.51% to 2.10%, again about a 39% relative improvement. A similar pattern appears on OBD, where SNIPS-R@20 in decile 10 increases from 1.9% to 2.7% (roughly +42% relative) and SNIPS-M@20 from 0.80% to 1.13%. These curves indicate that PTE-GT not only narrows the gap on popular items but also achieves the largest proportional gains on the coldest items, which are typically the most affected by exposure bias.

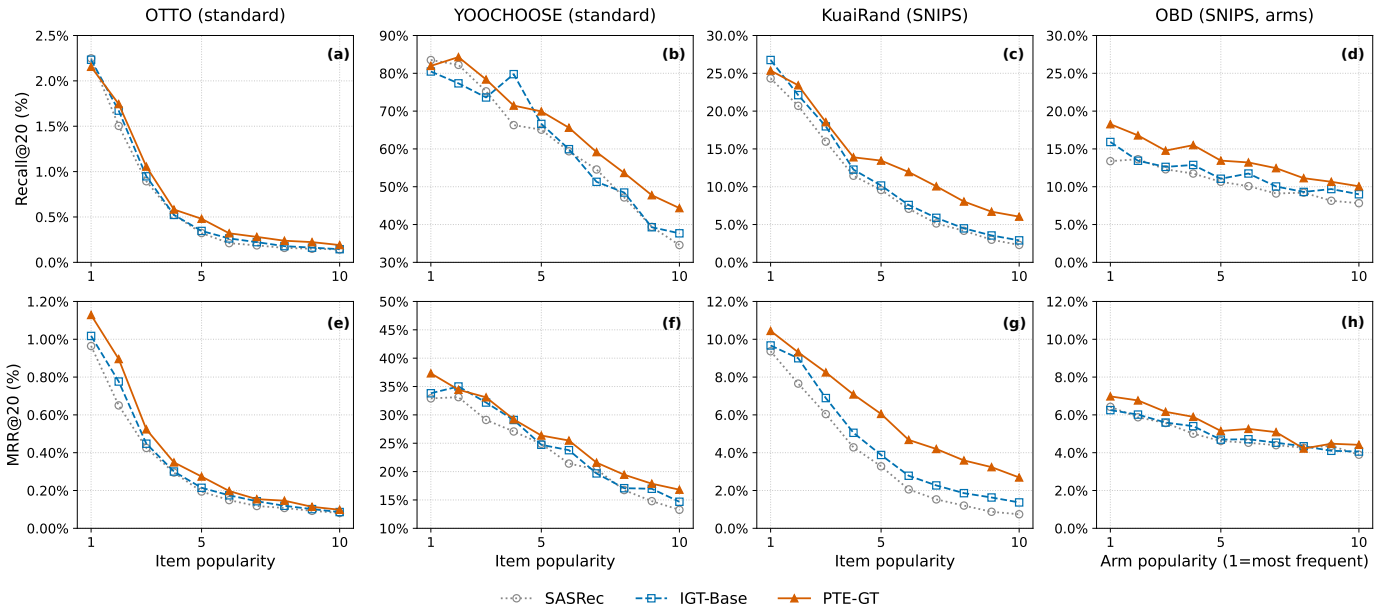


Figure 5. Comprehensive analysis of long-tail performance across item popularity deciles (1 = most popular, 10 = long-tail). (**Top row**) shows Recall@20. (**Bottom row**) shows MRR@20.

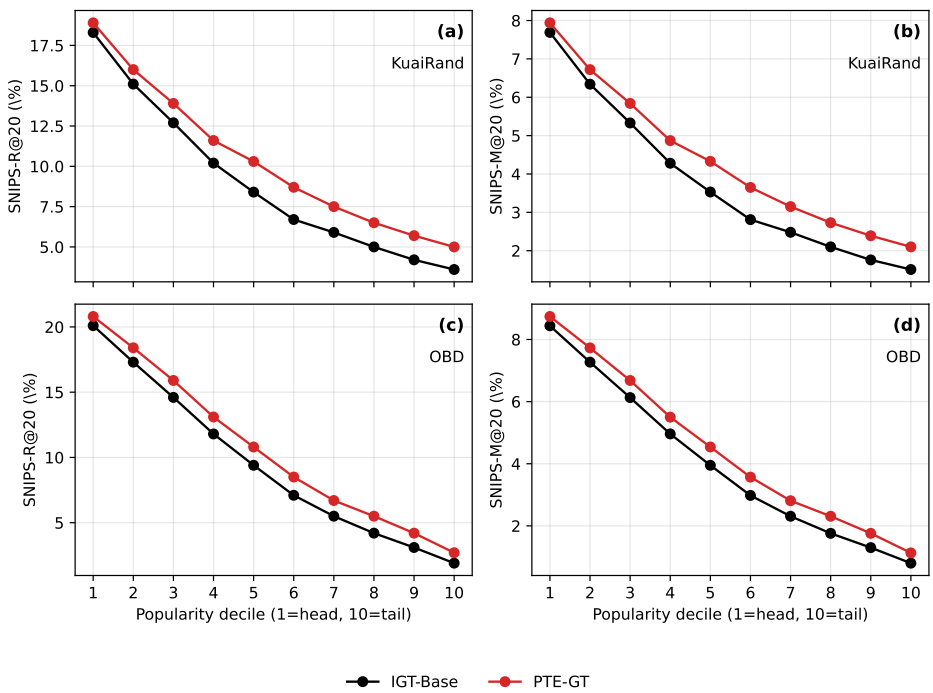
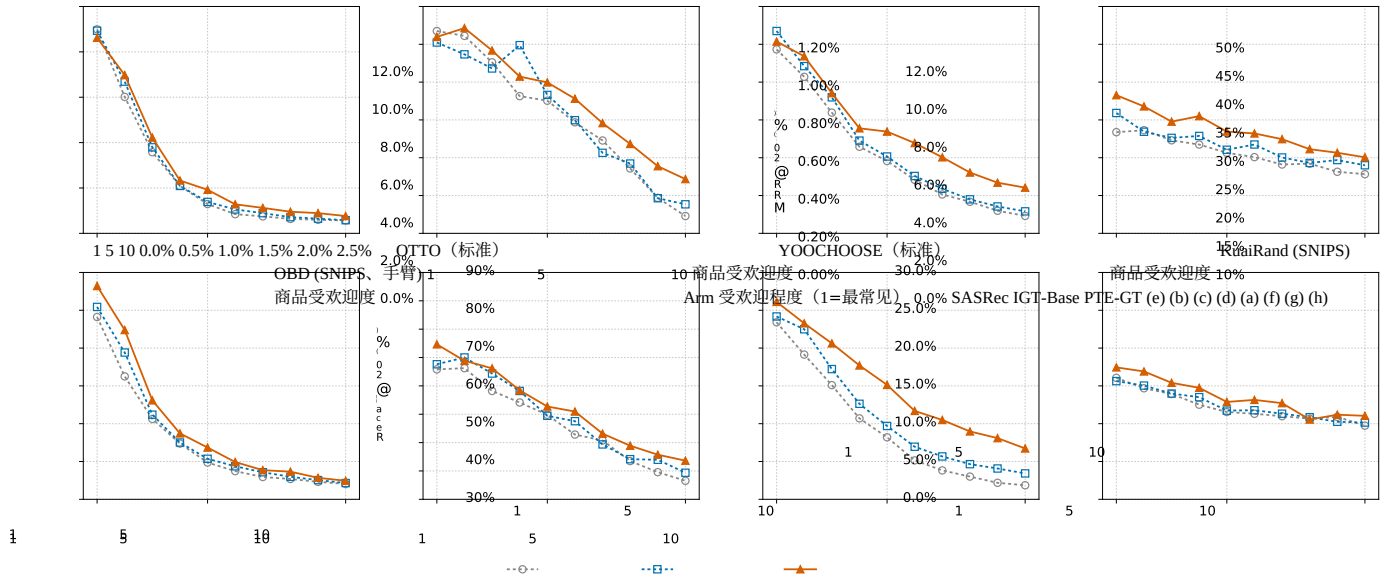
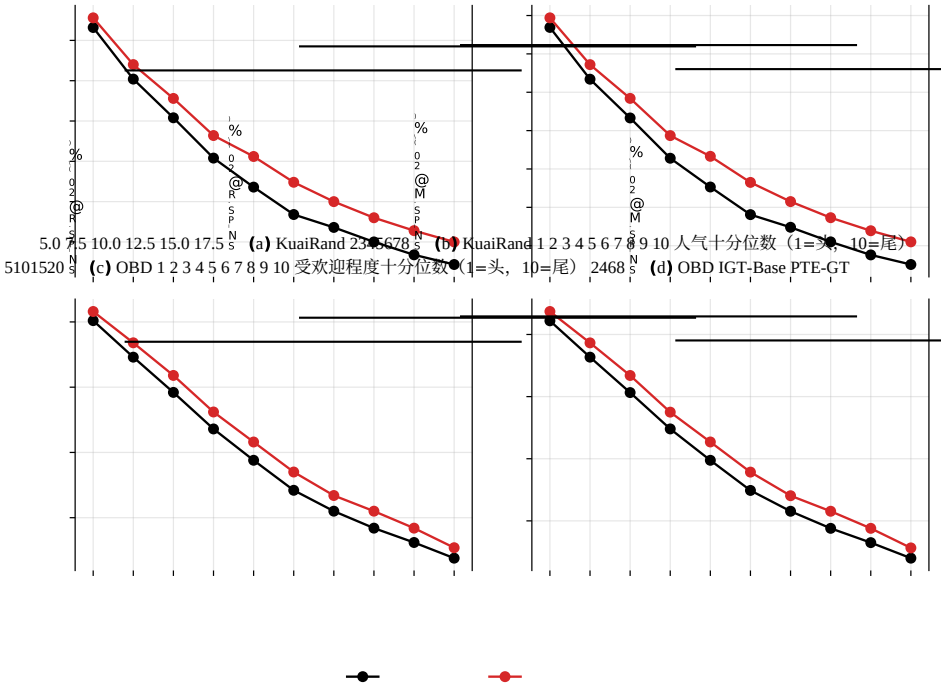


Figure 6. SNIPS-weighted long-tail performance by item popularity decile on KuaiRand (KuaiRand) and OBD (OBD). The figure reports SNIPS-R@20 and SNIPS-M@20 across popularity deciles.

 将 SNIPS-R@20 在头部十分位数 (1) 中从 18.3% 提高到 18.9%，但相对增益向尾部稳步增长，在底部 10% 项目中达到 +1.4 个百分点（3.6% 到 5.0%，大约为 +39%）。尾部相应的 SNIPS-M@20 从 1.51% 上升到 2.10%，同样相对提高了约 39%。 OBD 上也出现了类似的模式，其中十分之一中的 SNIPS-R@20 从 1.9% 增加到 2.7%（相对于 +42%），SNIPS-M@20 从 0.80% 增加到 1.13%。这些曲线表明，PTE-GT 不仅缩小了热门项目的差距，而且在最冷的项目上实现了最大的比例增益，这些项目通常受曝光偏差的影响最大。图 5. 跨商品流行度十分位数的长尾性能综合分析（1 = 最受欢迎，10 = 长尾）。 （顶行）显示 Recall@20。 （底行）显示 MRR@20。图 6. KuaiRand (KuaiRand) 和 OBD (OBD) 上按项目受欢迎程度十分位划分的 SNIPS 加权长尾表现。该图报告了 SNIPS-R@20 和 SNIPS-M@20 的受欢迎程度。 https://doi.org/10.3390/ electronics15010084





7. Ablation and Analysis

To deeply understand the contributions of each component in our proposed PTE-GT model, we conduct a series of detailed ablation studies. We primarily investigate two core modules: (1) PA (Propensity-Aware Learning), the SNIPS-based propensity-aware objective function; and (2) TC (Temporal Consistency), our view-free temporal consistency contrastive regularizer. Our baseline model is IGT-Base, which uses only the interval-aware graph transformer backbone and is trained with a standard cross-entropy loss.

7.1. Impact of Core Components

We evaluate different variants of our model on the KuaiRand-27K and OTTO datasets. KuaiRand-27K allows us to use SNIPS-weighted metrics to assess unbiased performance, while OTTO represents a standard (biased) large-scale benchmark. We evaluate four key metrics: (1) SNIPS-R@20 (unbiased accuracy), (2) ECE (Expected Calibration Error, lower is better), (3) Long-Tail R@20 (unbiased recall on the bottom 60% least popular items on KuaiRand), and (4) R@20 (standard biased recall on OTTO). Results are summarized in Table 6.

Table 6. Ablation study of PTE-GT components. IGT-Base is our backbone, PA applies the SNIPS objective, and TC adds temporal consistency regularization.

Model Variant	KuaiRand-27K (Unbiased)			OTTO (Standard)	
	SNIPS-R@20 (%)	ECE (%)	Long-Tail R@20 (%)	R@20 (%)	ECE (%)
(1) IGT-Base	10.88	6.45	4.93	47.92	5.12
(2) w/o PA (IGT-Base + TC)	11.05	6.23	5.11	48.31	4.95
(3) w/o TC (IGT-Base + PA)	12.65	3.33	7.81	48.82	3.12
(4) PTE-GT (Full Model)	13.05	3.11	8.15	49.13	2.85

The results in Table 6 clearly demonstrate the orthogonal contributions and synergistic effects of our two modules:

- PA is Key to Debiasing and Calibration: Comparing (1) IGT-Base and (3) w/o TC, adding only the PA module brings the largest performance leap. On KuaiRand, unbiased SNIPS-R@20 increased from 10.88% to 12.65%. More importantly, PA significantly improved model calibration, with ECE dropping from 6.45% to 3.33%, and long-tail recommendation capability (Long-Tail R@20 from 4.93% to 7.81%). This confirms that propensity weighting is the core mechanism for addressing exposure bias, recovering long-tail items, and making model confidence more reliable.
- TC Provides Robust Regularization: Comparing (3) w/o TC and (4) PTE-GT, adding the TC module on top of an already PA-enhanced model further improved SNIPS-R@20 from 12.65% to 13.05% and long-tail recall to 8.15%. Similarly, comparing (1) IGT-Base and (2) w/o PA, the TC module also brought moderate but consistent improvements to standard training (R@20 from 47.92% to 48.31%) and calibration, with ECE decreasing from 5.12% to 4.95%. This indicates that our temporal consistency contrastive loss acts as an effective regularizer, further enhancing generalization and robustness by encouraging the model to learn more time-invariant session representations. Figure 7 illustrates these trends visually: on KuaiRand, the largest absolute gains appear on long-tail SNIPS-R@20, whereas on OTTO the gains are smaller but consistently positive across all variants. These empirical gains are consistent with the general theory of consistency training on sequences [33,34], where enforcing agreement between multiple views of the same latent intent is expected to reduce representation variance and enhance robustness, especially under distributional shift and exposure bias.

 7. 消融和分析 为了深入了解我们提出的 PTE-GT 模型中每个组件的贡献，我们进行了一系列详细的消融研究。我们主要研究两个核心模块：（1）PA（倾向感知学习），基于SNIPS的倾向感知目标函数； (2) TC（时间一致性），我们的无视图时间一致性对比正则化器。我们的基线模型是 IGT-Base，它仅使用区间感知图转换器主干，并使用标准交叉熵损失进行训练。 *7.1. Impact of Core Components* 我们在 KuaiRand-27K 和 OTTO 数据集上评估模型的不同变体。 KuaiRand-27K 允许我们使用 SNIPS 加权指标来评估无偏差性能，而 OTTO 则代表标准（有偏差）的大规模基准。我们评估四个关键指标：(1) SNIPS-R@20（无偏准确度）、(2) ECE（预期校准误差，越低越好）、(3) 长尾 R@20（快兰德上底部 60% 最不受欢迎的项目的无偏召回）和 (4) R@20（OTTO 上的标准有偏召回）。结果总结于表 6 中。

Table 6. Ablation study of `PTE-GT` components. `IGT-Base` is our backbone, `PA` applies the SNIPS objective, and `TC` adds temporal consistency regularization. 模型变体 KuaiRand-27K（无偏） OTTO（标准） SNIPS-R@20 (%) ECE (%) 长尾 R@20 (%) R@20 (%) ECE (%) (1) IGT-Base 10.88 6.45 4.93 47.92 5.12 (2) 无 PA (IGT-Base + TC) 11.05 6.23 5.11 48.31 4.95 (3) 无 TC (IGT-Base + PA) 12.65 3.33 7.81 48.82 3.12 (4) PTE-GT（完整型号） 13.05 3.11 8.15 49.13 2.85 表 6 中的结果清楚地表明了我们两个模块的正交贡献和协同效应：

- PA 是去偏和校准的关键：比较 (1) IGT-Base 和 (3) w/o TC，仅添加 PA 模块可带来最大的性能飞跃。在 KuaiRand 上，无偏 SNIPS-R@20 从 10.88% 增加到 12.65%。更重要的是，PA显着改善了模型校准，ECE从6.45%下降到3.33%，长尾推荐能力（Long-Tail R@20从4.93%下降到7.81%）。这证实了倾向加权是解决暴露偏差、恢复长尾项目、使模型置信度更可靠的核心机制。

- TC 提供稳健的正则化：比较 (3) 无 TC 和 (4) PTE-GT，在已经 PA 增强的模型之上添加 TC 模块，进一步将 SNIPS-R@20 从 12.65% 提高到 13.05%，将长尾召回率提高到 8.15%。同样，比较 (1) IGT-Base 和 (2) w/o PA，TC 模块也为标准训练（R@20 从 47.92% 到 48.31%）和校准带来了适度但一致的改进，ECE 从 5.12% 下降到 4.95%。这表明我们的时间一致性对比损失充当了有效的正则化器，通过鼓励模型学习更多时间不变的会话表示来进一步增强泛化性和鲁棒性。图 7 直观地说明了这些趋势：在 KuaiRand 上，最大的绝对增益出现在长尾 SNIPS-R@20 上，而在 OTTO 上，增益较小，但在所有变体中始终为正值。这些经验收益与序列一致性训练的一般理论一致[33,34]，其中在同一潜在意图的多个视图之间强制达成一致有望减少表示方差并增强鲁棒性，特别是在分布偏移和暴露偏差的情况下。 https://doi.org/10.3390/ electronics15010084

- Biased Metrics Mask the True Gap: On the OTTO dataset, the standard R@20 improvement of the full model (49.13%) over the baseline (47.92%) appears modest (+2.5%). However, in the unbiased evaluation on KuaiRand, the relative improvement in SNIPS-R@20 is a striking 19.9% (from 10.88% to 13.05%). This highlights that evaluating models on standard (biased) metrics severely underestimates the true problem caused by exposure bias and the genuine value brought by our propensity-aware approach.

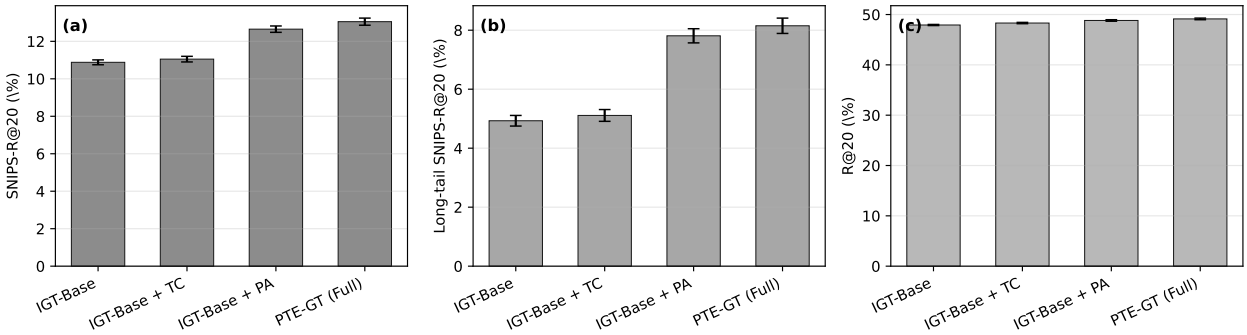


Figure 7. Component-wise ablation on KuaiRand and OTTO. Panel (**a**) shows mean SNIPS-R@20 on KuaiRand, panel (**b**) shows SNIPS-R@20 on the bottom 60% least popular items on KuaiRand (long-tail), and panel (**c**) shows standard R@20 on OTTO. Error bars indicate the standard deviation over five runs.

7.2. Sensitivity to SNIPS Clipping Threshold c

Our propensity-aware learning relies on the SNIPS estimator, where the clipping threshold c is a key hyperparameter. c controls the bias–variance trade-off: a small c clips large propensity weights, reducing variance but reintroducing bias; a large c allows for more extreme weights, reducing bias but increasing variance and training instability.

We train our `PTE-GT` model on the KuaiRand dataset while varying the value of c (from 1 to 100) and plot the results in Figure 8. This combination chart clearly illustrates the bias–variance trade-off.

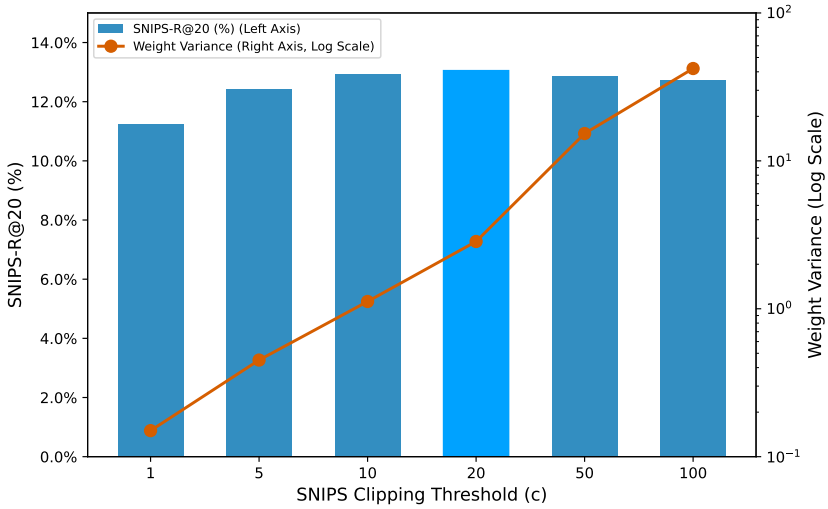
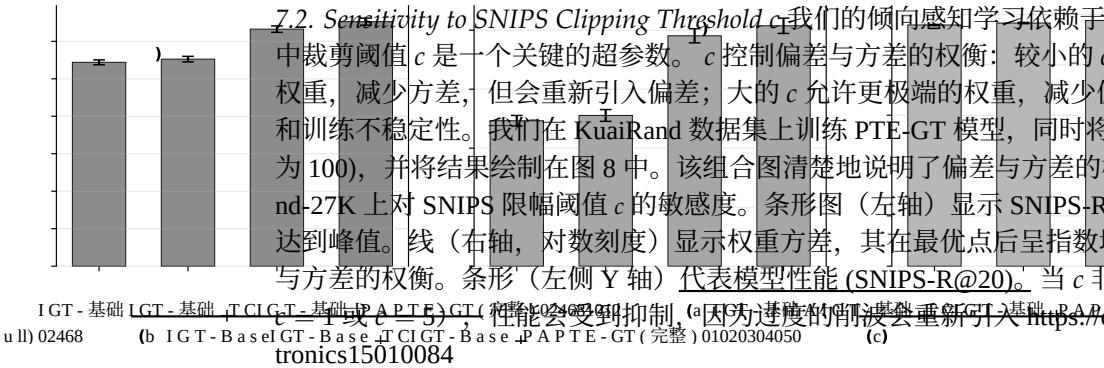


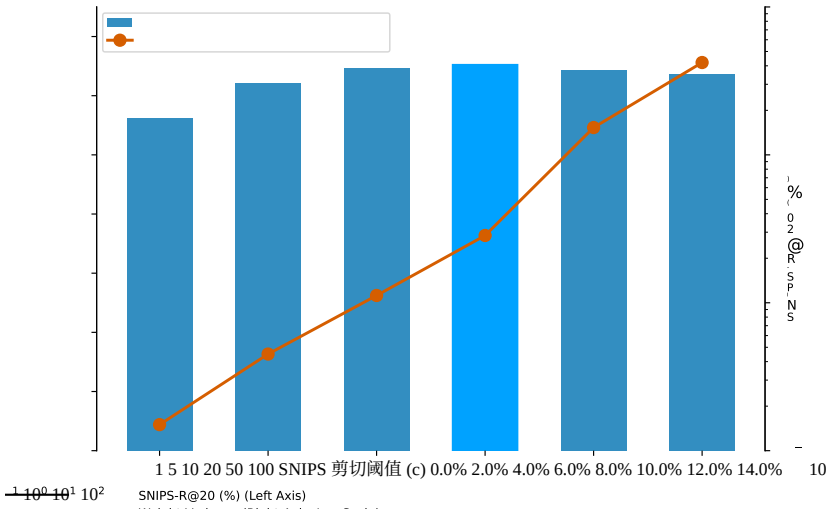
Figure 8. Sensitivity to the SNIPS clipping threshold c on KuaiRand-27K. The bars (left axis) show SNIPS-R@20, peaking at $c = 20$. The line (right axis, log scale) shows the weight variance, which increases exponentially after the optimal point, illustrating the bias–variance trade-off.

The bars (left Y-axis) represent the model performance (SNIPS-R@20). When c is very small (e.g., $c = 1$ or $c = 5$), performance is suppressed, as excessive clipping reintroduces

- 有偏差的指标掩盖了真正的差距：在 OTTO 数据集上，完整模型的标准 R@20 改进 (49.13%) 相对于基线 (47.92%) 似乎不大 (+2.5%)。然而，在快兰德的公正评估中，SNIPS-R@20 的相对改进达到了惊人的 19.9%（从 10.88% 到 13.05%）。这凸显了根据标准（有偏差的）指标评估模型严重低估了暴露偏差造成的真正问题以及我们的倾向感知方法带来的真正价值。图 7. KuaiRand 和 OTTO 上的按组件消融。图（a）显示了快兰德上的平均 SNIPS-R@20，图（b）显示了快兰德上最不受欢迎的 60% 商品（长尾）上的 SNIPS-R@20，图（c）显示了 OTTO 上的标准 R@20。误差线表示五次运行的标准偏差。

7.2. Sensitivity to SNIPS Clipping Threshold c 我们的倾向感知学习依赖于 SNIPS 估计器，其中裁剪阈值 c 是一个关键的超参数。c 控制偏差与方差的权衡：较小的 c 会削减较大的倾向权重，减少方差，但会重新引入偏差；大的 c 允许更极端的权重，减少偏差，但增加方差和训练不稳定性。我们在 KuaiRand 数据集上训练 PTE-GT 模型，同时将 c (的值从 1 更改为 100)，并将结果绘制在图 8 中。该组合图清楚地说明了偏差与方差的权衡。图 8. KuaiRand-27K 上对 SNIPS 限幅阈值 c 的敏感度。条形图（左轴）显示 SNIPS-R@20，在 $c = 20$ 处达到峰值。线（右轴，对数刻度）显示权重方差，其在最优点后呈指数增加，说明了偏差与方差的权衡。条形（左侧 Y 轴）代表模型性能 (SNIPS-R@20)。当 c 非常小时（例如，$c = 1$ 或 $c = 5$），性能会受到抑制，因为过度的削波会重新引入 https://doi.org/10.3390/electronics15010084





bias. Performance steadily improves as c increases, peaking at $c = 20$ (13.05%). Beyond this point ($c = 50, 100$), performance slightly degrades.

The line (right Y-axis, log scale) represents the SNIPS weight variance ($\mathcal{R}_{\text{var}}$). This variance remains low and stable for $c \leq 20$ but increases exponentially thereafter. This sharp rise in variance coincides with the performance degradation, as the extreme weights lead to an unstable training process. Therefore, we select $c = 20$ as the optimal trade-off for all our experiments.

To further quantify this bias–variance trade-off and to assess its robustness across datasets and propensity model capacities, Figure 9 extends the analysis to both KuaiRand and OBD. Figure 9a plots SNIPS-R@20 as a function of the clipping threshold $c \in \{2, 5, 10, 15, 20, 30, 50, 75, 100\}$ when using the LR-Full propensity model. On KuaiRand, performance increases steadily from 12.40% at $c = 2$ to a peak of 13.05% at $c = 20$, then declines slightly for larger thresholds (12.90% at $c = 100$). OBD exhibits a similar pattern, rising from 11.88% at $c = 2$ to 12.74% at $c = 20$ before plateauing. Figure 9b reports the corresponding weight variance on a logarithmic scale: for both datasets, $\text{Var}(w)$ remains close to one up to $c = 20$ but grows rapidly thereafter, reaching 3.00 on KuaiRand and 2.80 on OBD at $c = 100$. Figure 9c,d compare the three propensity models from Table 2—LR-Restrict, LR-Full, and MLP-2layer—in terms of AUC versus ECE and SNIPS-R@20 versus $\text{Var}(w)$. LR-Full attains almost the same SNIPS-R@20 as the MLP (13.05% vs. 13.09% on KuaiRand; 12.74% vs. 12.81% on OBD) while keeping $\text{Var}(w)$ much closer to LR-Restrict, placing it closest to the empirical Pareto frontier on both datasets.

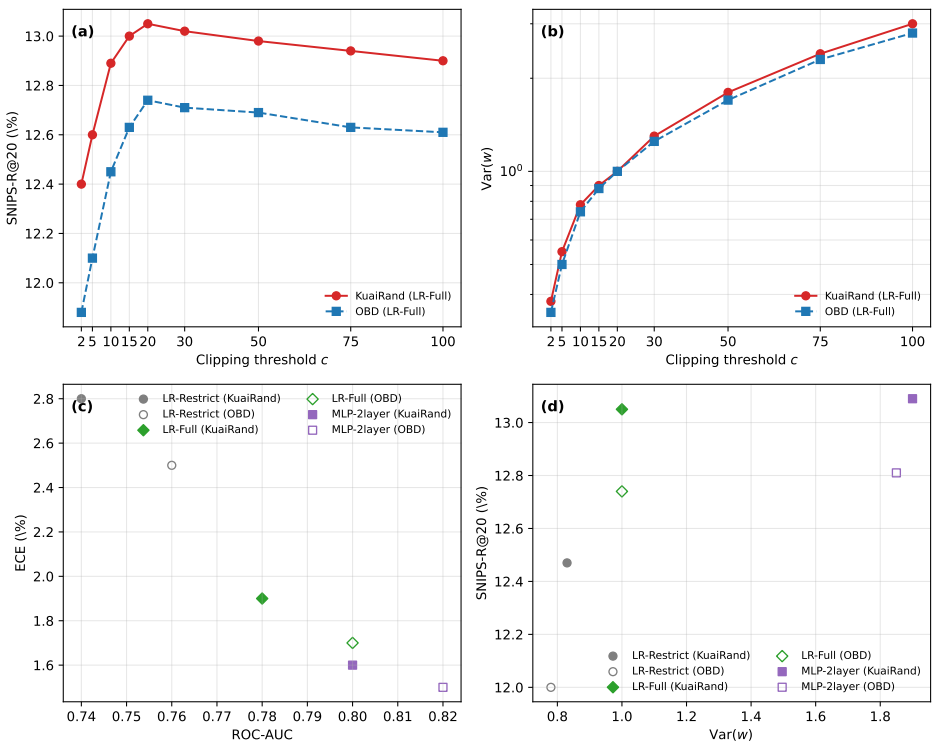


Figure 9. Bias–variance and propensity model analysis on KuaiRand and OBD. Panel (**a**) shows SNIPS-R@20 as a function of the clipping threshold c for KuaiRand and OBD using the LR-Full propensity model. Panel (**b**) plots the corresponding variance of the SNIPS weights on a logarithmic scale. Panel (**c**) compares ROC-AUC versus ECE for three propensity models (LR-Restrict, LR-Full, MLP-2layer) on both datasets. Panel (**d**) shows SNIPS-R@20 versus $\text{Var}(w)$.

On both KuaiRand and OBD, SNIPS-R@20 increases steadily as the clipping threshold c grows from 5 to 20 (e.g., 12.60% to 13.05% on KuaiRand; 12.10% to 12.74% on OBD) but starts to saturate and slightly degrade beyond $c = 20$, while the weight variance explodes

 偏差。随着 c 的增加，性能稳步提高，在 $c = 20$ 时达到峰值 (13.05%)。超过此点 ($c = 50, 100$)，性能会略有下降。线（右 Y 轴，对数刻度）表示 SNIPS 权重方差 ($\mathcal{R}_{\text{var}}$)。对于 $c \leq 20$，该方差保持较低且稳定，但此后呈指数增长。方差的急剧上升与性能下降同时发生，因为极端的权重会导致训练过程不稳定。因此，我们选择 $c = 20$ 作为我们所有实验的最佳权衡。为了进一步量化这种偏差-方差权衡并评估其在数据集和倾向模型能力方面的稳健性，图 9 将分析扩展到 KuaiRand 和 OBD。图 9a 将 SNIPS-R@20 绘制为使用 LR-Full 倾向模型时剪裁阈值 $c \in \{2, 5, 10, 15, 20, 30, 50, 75, 100\}$ 的函数。在 KuaiRand 上，性能从 $c = 2$ 时的 12.40% 稳步增长到 $c = 20$ 时的峰值 13.05%，然后在较大阈值时略有下降（$c = 100$ 时为 12.90%）。 OBD 也表现出类似的模式，从 $c = 2$ 时的 11.88% 上升到 $c = 20$ 时的 12.74%，然后达到稳定状态。图 9b报告了对数尺度上相应的权重方差：对于两个数据集，Var(w)在$c = 20$之前仍然接近1，但此后迅速增长，在$c = 100$时在KuaiRand上达到3.00，在OBD上达到2.80。图9c，d比较了表2中的三个倾向模型——LR-Restrict、LR-Full和MLP-2layer - 就 AUC 与 ECE 以及 SNIPS-R@20 与 Var(w) 而言。 LR-Full 获得了与 MLP 几乎相同的 SNIPS-R@20（KuaiRand 上为 13.05% vs. 13.09%；OBD 上为 12.74% vs. 12.81%），同时使 Var(w) 更接近 LR-Restrict，使其最接近两个数据集上的经验 Pareto 前沿。图 9. KuaiRand 和 OBD 的偏差-方差和倾向模型分析。图 (a) 显示 SNIPS-R@20 作为 KuaiRand 和 OBD 使用 LR-Full 倾向模型的裁剪阈值 c 的函数。图 (b) 在对数刻度上绘制了 SNIPS 权重的相应方差。图 (c) 比较了两个数据集上三种倾向模型（LR-Restrict、LR-Full、MLP-2layer）的 ROC-AUC 与 ECE。图 (d) 显示了 SNIPS-R@20 与 Var(w)。在 KuaiRand 和 OBD 上，SNIPS-R@20 随着裁剪阈值 c 从 5 增长到 20（例如，KuaiRand 上从 12.60% 到 13.05%；OBD 上从 12.10% 到 12.74%）稳步增加，但在超过 $c = 20$ 后开始饱和并略有下降，同时权重方差爆炸

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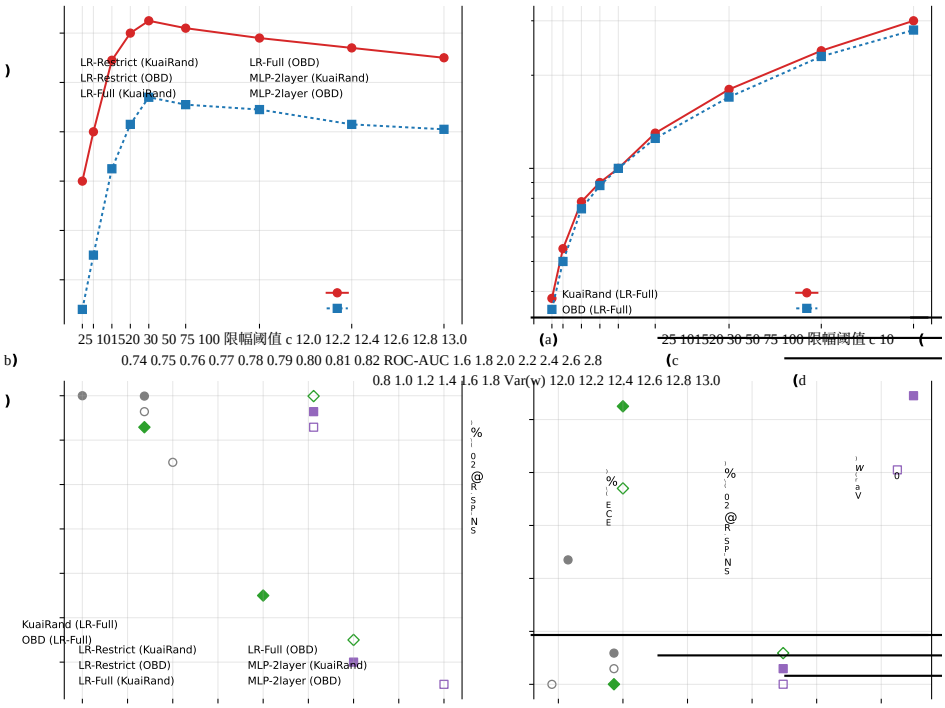


on a logarithmic scale (e.g., from 1.00 up to 3.00 on KuaiRand and 2.80 on OBD at $c = 100$). This confirms that our default choice $c = 20$ is robust across datasets with randomized and logged propensities.

7.3. Context Window Size on the Open Bandit Dataset

On OBD, we further examine how the impression-level context window size w affects performance for both PTE-GT and the SASRec-IPS baseline. Figure 10 reports SNIPS-R@20 and SNIPS-M@20 as functions of $w \in \{0, 3, 5, 8, 10, 15, 20, 30\}$. For PTE-GT, SNIPS-R@20 increases from 11.86% at $w = 0$ (no history) to a peak of 12.74% at $w = 10$, then slightly decreases to 12.68% and 12.60% at $w = 20$ and $w = 30$, respectively. SASRec-IPS exhibits a similar shape but remains consistently below PTE-GT (e.g., 10.85% to 11.48% to 11.40%). The SNIPS-M@20 curves follow the same pattern. Very short histories ($w = 0$ or $w = 5$) do not fully capture short-term preferences, while overly long histories ($w \geq 20$) introduce additional noise. A moderate window of $w = 10$ therefore offers the best trade-off on OBD and is used in all main experiments.

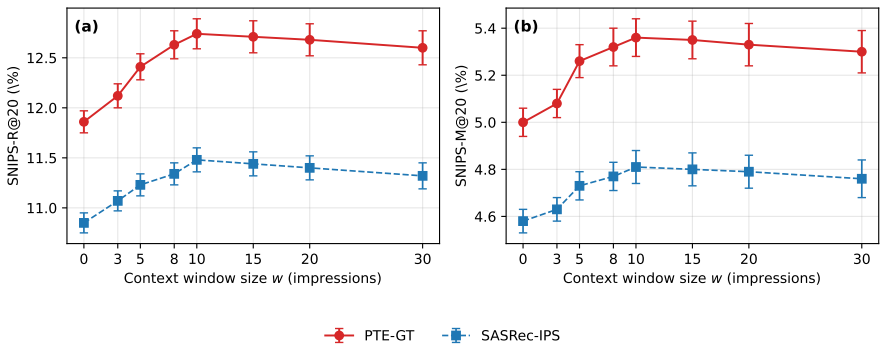


Figure 10. Effect of the context window size w on OBD. Panels (**a**,**b**) show SNIPS-R@20 and SNIPS-M@20, respectively, for PTE-GT and SASRec-IPS as w varies from 0 to 30 impressions. Points denote means over three random seeds and error bars denote one standard deviation.

Statistical significance results are summarized in Table 7.

Table 7. Statistical significance of PTE-GT compared to strong baselines (R@20 or SNIPS-R@20, five random seeds). Mean differences and confidence intervals are reported in percentage points (pp).

Dataset	Metric	Comparison	Mean Diff (pp)	95% CI (pp)	*p*-Value
OTTO	R@20	PTE-GT vs. TRON	+1.91	[+1.70, +2.10]	$<10^{-4}$
OTTO	R@20	PTE-GT vs. IGT-Base	+1.24	[+1.05, +1.43]	0.0003
YOOCHOOSE	R@20	PTE-GT vs. SR-GNN	+1.30	[+0.65, +1.95]	0.002
YOOCHOOSE	R@20	PTE-GT vs. IGT-Base	+0.89	[+0.05, +1.73]	0.041
KuaiRand	SNIPS-R@20	PTE-GT vs. IGT-Base	+2.17	[+1.96, +2.38]	$<10^{-4}$
OBD	SNIPS-R@20	PTE-GT vs. SASRec-IPS	+1.26	[+0.72, +1.80]	0.001
OBD	SNIPS-R@20	PTE-GT vs. DR-JL	+0.64	[+0.15, +1.13]	0.033

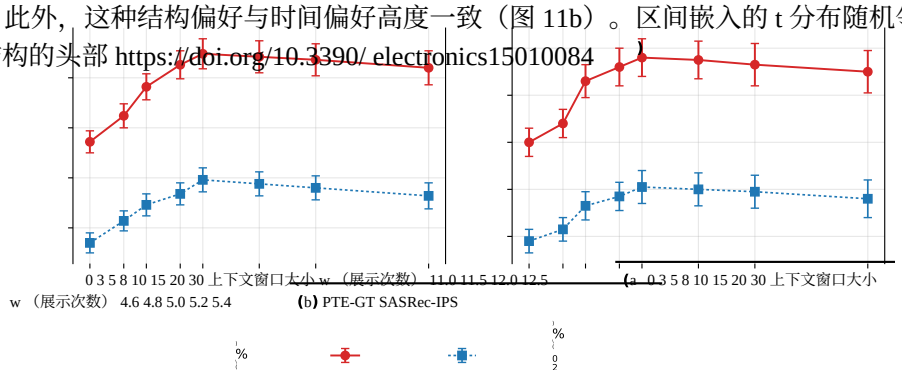
7.4. Qualitative Analysis of Interval-Aware Attention

To provide insight into our IGT backbone, we analyze its internal attention mechanism (Figure 11). We find clear evidence of head specialization. The model learns to distribute attention over distinct structural patterns Figure 11a, such as 'Immediate Forward' edges versus 'Far-Forward Shortcut' edges. Furthermore, this structural preference strongly aligns with temporal preferences Figure 11b. A t-distributed Stochastic Neighbor Embedding (t-SNE) projection of the interval embeddings shows that heads specializing in structural

采用对数刻度（例如，KuaiRand 上从 1.00 到 3.00，OBD 上 $c = 100$ 时为 2.80）。这证实了我们的默认选择 $c = 20$ 在具有随机和记录倾向的数据集中是稳健的。

7.3. Context Window Size on the Open Bandit Dataset 在 OBD 上，我们进一步研究展示级别上下文窗口大小 w 如何影响 PTE-GT 和 SASRec-IPS 基线的性能。图 10 将 SNIPS-R@20 和 SNIPS-M@20 报告为 $w \in \{0$、3、5、8、10、15、20、30$\}$ 的函数。对于 PTE-GT，SNIPS-R@20 从 $w = 0$（无历史）时的 11.86% 增加到 $w = 10$ 时的峰值 12.74%，然后分别在 $w = 20$ 和 $w = 30$ 时略微下降至 12.68% 和 12.60%。 SASRec-IPS 表现出类似的形状，但始终低于 PTE-GT（例如，10.85% 到 11.48% 到 11.40%）。 SNIPS-M@20 曲线遵循相同的模式。非常短的历史（$w = 0$ 或 $w = 5$）不能完全捕捉短期偏好，而过长的历史（$w \geq 20$）会引入额外的噪音。因此，$w = 10$ 的适中窗口提供了 OBD 的最佳权衡，并在所有主要实验中使用。图 10. 上下文窗口大小 w 对 OBD 的影响。面板 (a,b) 分别显示 PTE-GT 和 SASRec-IPS 的 SNIPS-R@20 和 SNIPS-M@20，因为 w 的展示次数从 0 到 30 不等。点表示三个随机种子的平均值，误差条表示一个标准差。统计显着性结果总结于表 7 中。 表 7. PTE-GT 与强基线（R@20 或 SNIPS-R@20，五个随机种子）相比的统计显着性。平均差异和置信区间以百分点 (pp) 为单位报告。数据集指标比较 平均差异 (pp) 95% CI (pp) p-值 OTTO R@20 PTE-GT 与 TRON +1.91 [+1.70, +2.10] $<10^{-4}$ OTTO R@20 PTE-GT 与 IGT-Base +1.24 [+1.05，+1.43] 0.0003 YOOCHOOSE R@20 PTE-GT 与 SR-GNN +1.30 [+0.65，+1.95] 0.002 YOOCHOOSE R@20 PTE-GT 与 IGT-Base +0.89 [+0.05, +1.73] 0.041 KuaiRand SNIPS-R@20 PTE-GT 与 IGT-Base +2.17 [+1.96, +2.38] $<10^{-4}$ OBD SNIPS-R@20 PTE-GT 与 SASRec-IPS +1.26 [+0.72, +1.80] 0.001 OBD SNIPS-R@20 PTE-GT 与 DR-JL +0.64 [+0.15, +1.13] 0.033

7.4. Qualitative Analysis of Interval-Aware Attention 为了深入了解我们的 IGT 主干，我们分析了其内部注意力机制（图 11）。我们发现了头部专业化的明显证据。该模型学习将注意力分配到不同的结构模式（图 11a），例如“立即向前”边缘与“远向快捷”边缘。此外，这种结构偏好与时间偏好高度一致（图 11b）。区间嵌入的 t 分布随机邻域嵌入 (t-SNE) 投影表明，专门研究结构的头部 https://doi.org/10.3390/ electronics15010084



shortcuts also learn to focus on long time intervals, while sequential heads focus on short-to-medium intervals.

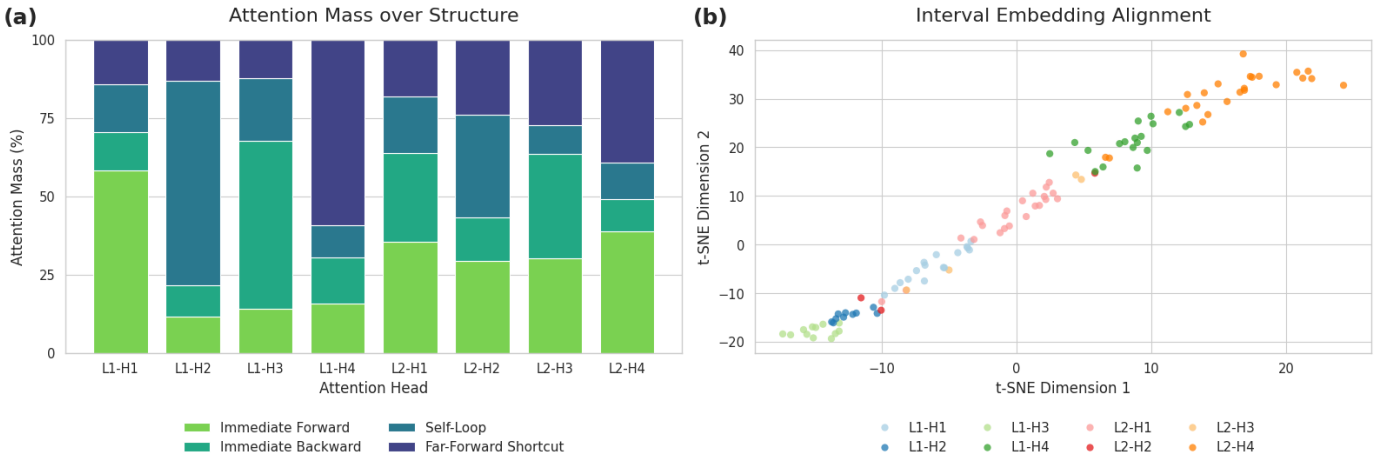


Figure 11. Qualitative analysis of the interval-aware attention mechanism. (**a**) Attention mass distribution over four structural edge types, revealing head specialization. (**b**) t-SNE projection of Fourier interval embeddings (Δt). Colors show the dominant attention head, demonstrating a clear alignment between structural preference (**a**) and temporal preference (**b**).

This spatio-temporal synergy is exemplified by Head 'L1-H4', which dedicates its mass to 'Far-Forward Shortcut' edges (59.2%) and concurrently dominates the long-interval region of the time manifold (orange cluster). In contrast, 'L1-H1' acts as a sequential processor (58.3% on 'Immediate Forward'). This confirms our model learns to find unified patterns, such as "long-term, time-delayed dependencies," rather than treating time and structure as independent features.

7.5. Comparison of Contrastive Learning Strategies

Finally, we validate the effectiveness of our proposed view-free temporal consistency regularizer `TC` by comparing it against a more standard graph augmentation-based contrastive learning method. We create a variant named `PTE-GT (GraphAug)`, which replaces `TC` with a common graph CL strategy (i.e., creating two stochastic augmentations of the session graph, such as 20% node dropping and 20% edge perturbation, and then maximizing the consistency between the two augmented views). Unbiased evaluation results on KuaiRand show that `PTE-GT (GraphAug)` achieves a SNIPS-R@20 of 12.98%, which is slightly lower than our standard `PTE-GT` model's 13.05%. However, in terms of computational cost, `PTE-GT (GraphAug)` introduces significant overhead: due to the need to construct and encode two separate, augmented graphs for each session, the training time per epoch increases by approximately 34%. In contrast, our proposed `TC` module, which operates only on the encoded representations, adds only about 8% training overhead. This comparison confirms that our designed view-free temporal consistency strategy is not only more computationally efficient but also more effective (at least compared to standard graph augmentation strategies), providing an effective and low-cost regularization for the model.

8. Case Study: Structure-Aware Evidence Flow to the Hit Item

To make the graph-specific reasoning of our model tangible, we visualize a single OTTO-style session using a Sankey diagram that aggregates evidence flows from upstream nodes to the top-1 hit item L. Edge width encodes a composite evidence weight

$$W_{ij} = \Big(\sum_{h=1}^{H} u_h \, \alpha_{ij}^{(h)} \Big) \cdot A_{ij} \cdot \exp\big(-\eta \, \Delta t_{ij} \big), \tag{14}$$

图 11. 间隔感知注意力机制的定性分析。 (a) 四种结构边缘类型上的注意力质量分布，揭示了头部的专业化。 (b) 傅立叶区间嵌入 (Δt) 的 t-SNE 投影。颜色显示了占主导地位的注意力头，表明结构偏好 (a) 和时间偏好 (b) 之间存在明显的一致性。这种时空协同作用的例子是头“L1-H4”，它将其质量专用于“远前捷径”边缘（59.2%），同时主导时间流形的长间隔区域（橙色簇）。相比之下，“L1-H1”充当顺序处理器（“立即转发”为 58.3%）。这证实了我们的模型学会寻找统一的模式，例如“长期的、延时的依赖关系”，而不是将时间和结构视为独立的特征。 *7.5. Comparison of Contrastive Learning Strategies*最后，我们通过将其与更标准的基于图增强的对比学习方法进行比较，验证了我们提出的无视图时间一致性正则化器 TC 的有效性。我们创建了一个名为 PTE-GT (GraphAug) 的变体，它用常见的图 CL 策略取代了 TC（即，创建会话图的两个随机增强，例如 20% 的节点丢弃和 20% 的边缘扰动，然后最大化两个增强视图之间的一致性）。 KuaiRand 上的无偏评估结果显示，PTE-GT (GraphAug) 的 SNIPS-R@20 为 12.98%，略低于我们的标准 PTE-GT 模型的 13.05%。然而，就计算成本而言，PTE-GT（GraphAug）引入了显着的开销：由于需要为每个会话构建和编码两个独立的增强图，因此每个时期的训练时间增加了大约 34%。相比之下，我们提出的 TC 模块仅对编码表示进行操作，仅增加了约 8% 的训练开销。这一比较证实了我们设计的无视图时间一致性策略不仅计算效率更高，而且更有效（至少与标准图增强策略相比），为模型提供了有效且低成本的正则化。 8. 案例研究：到命中项的结构感知证据流 为了使我们模型的特定于图的推理变得具体，我们使用 Sankey 图可视化单个 OTTO 式会话，该图聚合从上游节点到 top-1 命中项 L 的证据流。边缘宽度编码复合证据权重 $W_{ij} = \left(\sum_{h=1}^{H} u_h \alpha_{ij}^{(h)}\right) \cdot A_{ij} \cdot \exp(-\eta \Delta t_{ij})$, (14)

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combining head-weighted graph attention $\alpha_{ij}^{(h)}$, edge attribution A_{ij} (Integrated Gradients), and a temporal decay governed by η. For each candidate path $\mathcal{P}$ to L, we summarize evidence by the geometric mean $\mathrm{PathW}(\mathcal{P})$ of $\{W_e\}_{e\in\mathcal{P}}$ and normalize across the top evidence paths to obtain flow proportions.

The session is $A(0\text{ s}) \to B(18) \to C(41) \to D(65) \to E(71) \to F(96) \to G(131) \to H(145) \to L(158)$, with two non-sequential (shortcut) edges: $C \to F$ ($\Delta t = 55$ s) and $B \to E$ ($\Delta t = 53$ s). The four highest-evidence paths to L are as follows:

- P4: $B \to C \to F \to H \to L$ (30.2%)
- P1: $A \to B \to C \to F \to G \to H \to L$ (28.5%)
- P2: $A \to B \to E \to F \to G \to H \to L$ (26.6%)
- P3: $A \to B \to C \to D \to E \to F \to G \to H \to L$ (14.7%)

These flows sum to 100% by construction. Notably, $C \to F$ and $B \to E$ carry the majority of total evidence via P4/P1/P2 (85.3%), while the purely sequential route P3 contributes only 14.7%.

Figure 12 visualizes this session.

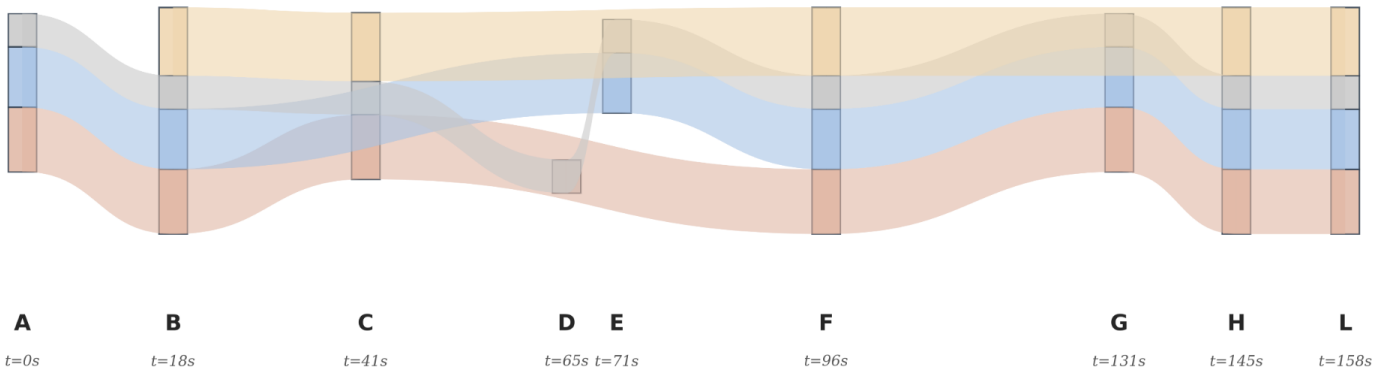


Figure 12. Case: Session flow Sankey with evidence weights. Evidence concentrates on time-salient shortcut edges ($C \to F$, $B \to E$), forming dominant paths to the hit item L (P4: 30.2%, P1: 28.5%, P2: 26.6%), while the purely sequential path (P3: 14.7%) is comparatively weaker. This highlights non-sequential, structure-aware reasoning afforded by the graph transformer.

The concentration of evidence on shortcut edges, despite longer temporal gaps, indicates that the model leverages graph connectivity beyond strict sequence order. This structure-aware behavior is precisely where the graph transformer outperforms sequential baselines.

9. Discussion and Conclusions

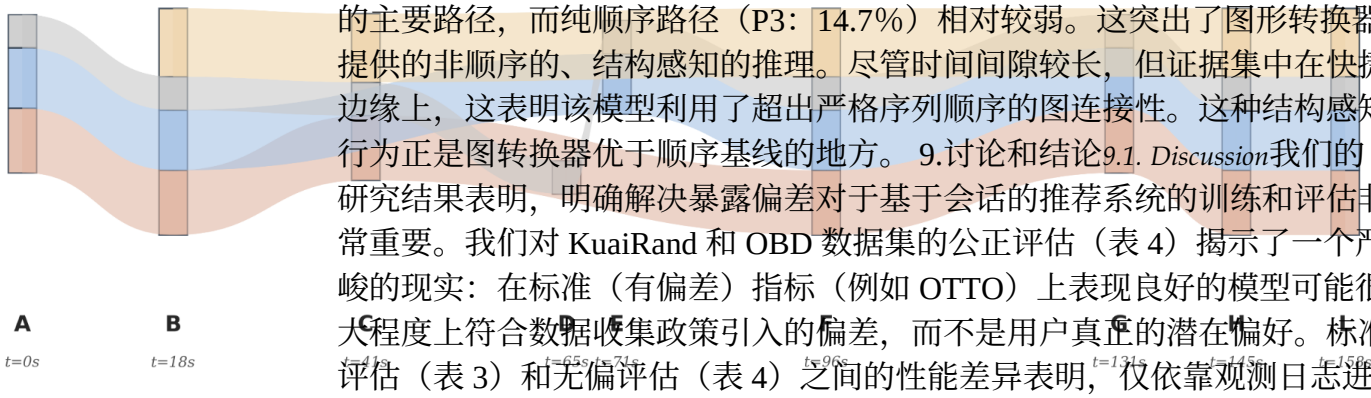
9.1. Discussion

Our findings suggest that explicitly addressing exposure bias is important in the training and evaluation of session-based recommender systems. Our unbiased evaluation on the KuaiRand and OBD datasets (Table 4) reveals a stark reality: models that perform well on standard (biased) metrics (e.g., on OTTO) may largely be fitting the biases introduced by the data collection policy, rather than the users' true underlying preferences. The disparity in performance between standard evaluation (Table 3) and unbiased evaluation (Table 4) indicates that relying solely on observational logs for model assessment can be misleading, especially when long-tail behavior is of interest.

Our framework, PTE-GT, addresses this challenge through two orthogonal and complementary modules. First, PA is core to achieving debiasing. By adopting the SNIPS estimator, the model is forced to re-weight its loss function, increasing its focus on items that were clicked despite having low exposure rates (i.e., low propensity scores). As shown

 结合了头部加权图注意力 $\alpha_{ij}^{(h)}$ 、边缘归因 A_{ij} (积分梯度) 和由 η 控制的时间衰减。对于每个候选路径 $\mathcal{P}$ 到 L，我们通过 $\{W_e\}_{e\in\mathcal{P}}$ 的几何平均 PathW($\mathcal{P}$) 总结证据，并对顶部证据路径进行标准化以获得流量比例。会话为 $A(0\text{ s}) \to B(18) \to C(41) \to D(65) \to E(71) \to F(96) \to G(131) \to H(145) \to L(158)$ ，有两个非顺序（快捷）边：$C \to F$ ($\Delta t = 55$ s) 和 $B \to E$ ($\Delta t = 53$ 秒)。通往 L 的四个最明显的路径如下：

- P4：$B \to C \to F \to H \to L$ (30.2%)
- P1：$A \to B \to C \to F \to G \to H \to L$ (28.5%)
- P2：$A \to B \to E \to F \to G \to H \to L$ (26.6%)
- P3：$A \to B \to C \to D \to E \to F \to G \to H \to L$ (14.7%) 这些流量通过构建总计为100%。值得注意的是，$C \to F$和$B \to E$通过P4/P1/P2携带了大部分证据（85.3%），而纯粹的顺序途径P3仅贡献了14.7%。图 12 直观地展示了该会话。图 12. 案例：带有证据权重的会话流 Sankey。证据集中在时间显着的快捷边缘（$C \to F$，$B \to E$）上，形成了命中项L (P4：30.2%，P1：28.5%，P2：26.6%) 的主要路径，而纯顺序路径（P3：14.7%）相对较弱。这突出了图形转换器提供的非顺序的、结构感知的推理。尽管时间间隙较长，但证据集中在快捷边缘上，这表明该模型利用了超出严格序列顺序的图连接性。这种结构感知行为正是图转换器优于顺序基线的地方。 9.讨论和结论*9.1. Discussion*我们的研究结果表明，明确解决暴露偏差对于基于会话的推荐系统的训练和评估非常重要。我们对 KuaiRand 和 OBD 数据集的公正评估（表 4）揭示了一个严峻的现实：在标准（有偏差）指标（例如 OTTO）上表现良好的模型可能很大程度上符合数据收集政策引入的偏差，而不是用户真正的潜在偏好。标准评估（表 3）和无偏评估（表 4）之间的性能差异表明，仅依靠观测日志进行模型评估可能会产生误导，特别是当关注长尾行为时。我们的框架 PTE-GT 通过两个正交且互补的模块来应对这一挑战。首先，PA 是实现去偏置的核心。通过采用 SNIPS 估计器，该模型被迫重新加权其损失函数，增加对尽管曝光率较低（即倾向得分较低）但仍被点击的项目的关注。如图所示



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in the ablation study (Table 6), PA is the primary driver for improving long-tail recall (e.g., +59% in Long-Tail R@20) and calibration (e.g., >48% reduction in ECE). This confirms our hypothesis that directly optimizing an unbiased risk estimate is the fundamental way to correct the model's systematic bias towards popular items.

Second, the view-free TC regularizer provides a lightweight yet efficient solution to enhance representation robustness. Unlike relying on expensive graph augmentation strategies (which, as our analysis showed, add >35% training overhead), our prefix–suffix alignment strategy is computationally inexpensive (adding only 8% overhead) while still delivering consistent performance gains (Table 6). It improves generalization by encouraging the model to learn representations that are insensitive to minor temporal perturbations within a session, serving as a valuable addition on top of the primary debiasing already performed by the PA module.

However, our method also has limitations, which point to directions for future work.The accuracy of the propensity model in this study is critical. Although our default deployment uses the calibrated LR-Full model and we empirically compared it against both a more restricted logistic regression and a two-layer MLP in Table 2, more complex non-linear models can provide slightly higher AUC at the cost of substantially larger weight variance as visualized in Figure 9. Exploring this bias–variance trade-off, as well as investigating more advanced variance reduction techniques (beyond SNIPS clipping), would be a valuable research direction.

In addition to classical IPS-based baselines, we also compared PTE-GT against a CausalRec-style front-door debiasing model and a doubly robust variant (DR-JL) built on top of SASRec. These causal/DR baselines consistently outperform plain IPS, but PTE-GT still achieves the best SNIPS-weighted accuracy on both KuaiRand and OBD, reinforcing the benefit of combining an interval-aware graph transformer with a carefully calibrated SNIPS objective and temporal consistency regularization.

9.2. Conclusions

In this paper, we proposed PTE-GT, a debiased graph transformer framework for session-based recommendation. Our model integrates two key training-time modules onto an advanced interval-aware graph transformer backbone: a propensity-aware objective function based on SNIPS to correct for exposure bias; and a lightweight, view-free temporal consistency contrastive regularizer to enhance the robustness of session representations.

Through a comprehensive evaluation on three public session-based benchmarks and one public logged bandit dataset (OBD), we demonstrated that our approach not only outperforms baselines on standard recommendation metrics but, more importantly, achieves significant performance leaps under an unbiased evaluation protocol enabled by randomized exposure or logged propensities. Our model exhibits superior performance in terms of recommendation accuracy, calibration, and long-tail item discovery. This work provides a practical and effective pathway toward building more reliable and fair session-based recommender systems in bias-laden real-world environments.

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Data Availability Statement: All public datasets used in this study are openly available: KuaiRand (available online: https://kuairand.com/ (accessed on 13 November 2025)); OTTO (available online: https://doi.org/10.34740/KAGGLE/DSV/4991874 (accessed on 13 November 2025)); YOOCHOOSE (available online: https://recsys.acm.org/recsys15/challenge/ (accessed on 13 November 2025)); and the Open Bandit Dataset (available online: https://research.zozo.com/data.html (accessed on 13 November 2025)).

Conflicts of Interest: The authors declare no conflicts of interest.

Notation

Notation	Description
S, G	A user session and its corresponding graph representation
x_i, t_i	An item interaction and its timestamp
T	The length (number of items) of a session
h_i, s	Node (item) representation and final session representation
d	Dimension of embedding vectors
L, H	Number of layers and attention heads in the transformer
Δt	Time interval feature between two events
ℓ_θ	The base cross-entropy loss function
$\hat{p}_n$	The estimated propensity score (exposure probability)
w_n, c	SNIPS weight and its clipping threshold
$\mathcal{J}_{\text{sup}}$	The propensity-aware SNIPS supervised objective
$\mathcal{L}_{\text{con}}$	The temporal consistency contrastive loss
τ	Temperature parameter for contrastive loss

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Notation	Description	Notation	Description
S, G	A user session and its corresponding graph representation		
x_i, t_i	An item interaction and its timestamp	T	The length (number of items) of a session
h_i, s	Node (item) representation and final session representation	d	Dimension of embedding vectors
L, H	Number of layers and attention heads in the transformer	Δt	Time interval feature between two events
ℓ_θ	The base cross-entropy loss function	$\hat{p}_n$	The estimated propensity score (exposure probability)
w_n, c	SNIPS weight and its clipping threshold	$\mathcal{J}_{\text{sup}}$	The propensity-aware SNIPS supervised objective
$\mathcal{L}_{\text{con}}$	The temporal consistency contrastive loss	τ	Temperature parameter for contrastive loss

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